

A Casimir–Secular Electroweak Mass Ratio and Its String/Kaluza–Klein Interpretation Problem

Alejandro Rivero¹

¹*EUPT, Universidad de Zaragoza*

(Dated: May 22, 2026)

Abstract

The Rivero–de Vries electroweak observation is formulated as a two-branch secular problem built from $J = s(s+1)$, with roots of $x^2 + Jx - J = 0$. The positive-root quotient $1 - x_+(3/4)/x_+(2) = 0.22310132\dots$ is used as a descriptive spectral clue near the W/Z complex-pole weak-angle ratio, with the low-energy pole spectrum taken as the working placement hypothesis. The negative branch belongs to the same diagonalization and is treated as a candidate scalar/order-parameter datum whose physical meaning requires a gauge-invariant scalar functional. Endpoint, interval Kaluza–Klein, and localized G_2 settings supply source-theory addresses for a two-channel kernel; the central theorem targets are the ordered assignment $(J_W, J_Z) = (3/4, 2)$, the derivation of the DeVries kernel entries, and the pole-matched scalar interpretation of the partner branch.

I. INTRODUCTION

The electroweak sector contains a dimensionless pole-spectrum ratio,

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}, \quad (1)$$

which is fixed experimentally by the pole positions of the charged and neutral weak vector bosons. In the Standard Model at tree level the same ratio is written as an angular datum in coupling space, while the Higgs vacuum expectation value fixes the common radial scale of the vector masses. This separation between projective and radial data gives the working language of the present paper.

The Rivero–de Vries observation starts from a mass operator built from Poincare Casimir data and produces a dimensionless two-branch equation. In the normalized notation used below, the equation is

$$x^2 + Jx - J = 0, \quad J = s(s+1). \quad (2)$$

The positive branch gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)} = 0.22310132\dots \quad (3)$$

The original note emphasized the proximity of this number to the mass-shell weak angle and motivated the operator through Casimir invariants and high-spin behavior [1]. In this paper Eq. (3) is used as a descriptive spectral clue. The central question asks which physical

object carries the square root in Eq. (2), and why that object is read at the low-energy W/Z pole ratio.

This placement question is the conceptual core. A standard grand-unified lineage uses representation theory to obtain a clean weak-angle value at a high scale. In $SU(5)$, and in $Spin(10)$ with the usual fermion content, the famous group-theoretic value is $\sin^2 \theta_W = 3/8$; contact with the observed low-energy angle is then made through running and thresholds [20]. The Rivero–de Vries construction places the scale burden on the pole-mass ratio itself. Therefore the manuscript must explain why a spectral datum should attach to the pole spectrum. A unification-scale boundary condition would define a different reading of the same algebra.

The string-theoretic setting uses the older dual-model and Regge ambition as part of the physical context. Open strings, endpoint data, Kaluza–Klein spectra, and boundary conditions are treated as possible mechanisms for organizing low-energy particle structure. String theory also incorporates Kaluza–Klein ideas, grand-unified representations, and supersymmetric structure in transmuted form [3]. The physical question is whether those tools can produce an electroweak secular datum at the observed pole spectrum. The numerical clue remains a clue until that mechanism is written.

The construction has four layers. The first layer is algebraic: a two-channel secular equation has roots whose ratio is fixed by the Casimir variable. The second layer is electroweak: the W/Z mass ratio is a projective datum along the broken-to-unbroken ray of the gauge-Higgs system. The third layer is spectral: the square root must be assigned to a pole, running, compactification, or boundary-condition object. The fourth layer is dynamical: strings, branes, Kaluza–Klein boundary conditions, or singular G_2 geometries may generate the secular equation or its branch structure.

The logical structure used below separates assumptions from theorem targets:

- $A1 : Q(J)$ is the algebraic two-channel object,
- $A2 : s_{\text{pole}}^2$ is read with complex pole masses,
- $A3 : (J_W, J_Z) = (3/4, 2)$ is the ordered electroweak assignment,
- $T1 : \text{derive } A3 \text{ from gauge-Higgs or source data,}$
- $T2 : \text{derive the route kernel entries from a source theory,}$
- $T3 : \text{map } x_- \text{ to a gauge-invariant scalar functional.}$

The algebraic calculation is explicit. The pole-mass comparison is a descriptive clue whose

source audit remains a separate task. The assignment of $J = 3/4$ to the charged W comparison and $J = 2$ to the neutral Z comparison is the central open analytical calculation. The negative branch is retained because it may encode the scalar/order-parameter side of electroweak symmetry breaking.

The paper is organized as follows. Section II gives the secular operator and its roots. Section III defines the pole observable and the placement problem. Section IV formulates the projective electroweak interpretation. Section V records the negative branch and its possible Higgs-scale role. Sections VI–XIII develop the string, endpoint, Kaluza–Klein, and G_2 mechanism tests in a common two-channel language. Sections XIV and XV delimit the flavor and global-gauge-group issues. Section XVI gives the conclusion and analytical status.

II. CASIMIR-SECULAR MASS OPERATOR

Let $J = s(s + 1)$ be the spin-Casimir eigenvalue. The normalized two-channel operator used in this manuscript is

$$\frac{Q(J)}{\mu^2} = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}. \quad (4)$$

Its characteristic equation is

$$\det \left(xI - \frac{Q(J)}{\mu^2} \right) = x^2 + Jx - J = 0. \quad (5)$$

The two roots are

$$x_{\pm}(J) = \frac{-J \pm \sqrt{J^2 + 4J}}{2}. \quad (6)$$

They obey

$$x_+(J) + x_-(J) = -J, \quad x_+(J)x_-(J) = -J. \quad (7)$$

For $J > 0$, the positive branch is positive and the negative branch is negative.

Historically, the electroweak number is obtained by sampling the positive branch at the two Casimir values

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (8)$$

This gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)}. \quad (9)$$

The expression is dimensionless; the overall scale μ cancels. The cancellation is essential for an electroweak interpretation because Eq. (9) can then be compared with a projective W/Z mass ratio. In the electroweak reading developed here, $J_H = 3/4$ is the Higgs/order-parameter doublet invariant and $J_{\text{adj}} = 2$ is the adjoint/current invariant. The ordered use of these two samples in the W/Z quotient remains the central derivation problem.

The relation to the original Rivero operator is obtained by restoring the Poincare Casimir eigenvalues $C_1 = m^2$ and $C_2 = -m^2 s(s+1)$. In the original construction, a mass-squared operator is formed from combinations of Casimir invariants, with a high-spin condition designed to preserve the asymptotic mass behavior relevant for Regge trajectories [1]. Equation (4) is the reduced dimensionless representative of that branch problem.

A. What the algebra fixes

The operator in Eq. (4) fixes three algebraic facts. First, the trace is $-J$. Second, the determinant is $-J$. Third, the two eigenvalues are tied by the same invariant:

$$\text{tr} \frac{Q(J)}{\mu^2} = -J, \quad \det \frac{Q(J)}{\mu^2} = -J. \quad (10)$$

Thus a physical derivation has to explain a simultaneous trace and determinant assignment. A model that gives only a square-root branch, or only a J -dependent diagonal term, leaves the DeVries construction incomplete. The required two-channel matrix has the schematic form

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = J, \quad \tau_J = J. \quad (11)$$

This is the compact target for the endpoint, Kaluza–Klein, and G_2 sections.

The square-root radical is then the discriminant of this two-channel problem:

$$\Delta_J = J^2 + 4J. \quad (12)$$

The radical has to attach to a physical object. The radical-placement section lists the main options. This paper follows the pole-spectrum placement because the positive and negative branches are interpreted as a single low-energy spectral clue.

B. Minimal assumptions and theorem target

A complete derivation should state the minimal assumptions that select Eq. (4). The current minimal list is:

1. dependence on available Poincare or internal invariants;
2. dimensional consistency after restoring the mass scale μ ;
3. a two-branch secular structure;
4. preservation of the high-spin or Regge asymptotic datum in the originating problem;
5. a trace/determinant condition equivalent to Eq. (10).

These assumptions define the uniqueness target. The theorem to prove would read: within a stated class of two-channel operators built from the allowed invariants, the high-spin condition and trace/determinant constraints select Eq. (4) up to similarity transformations and normalization.

This theorem would also clarify the status of the negative branch. Since $x_-(J)$ is fixed by the same trace and determinant as $x_+(J)$, a derivation of Eq. (4) automatically derives both branches. The electroweak interpretation can then ask which fields or boundary data realize the two eigenvectors.

III. POLE OBSERVABLE AND ELECTROWEAK SCHEME SEPARATION

The observable used for comparison is the pole-mass ratio

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (13)$$

This ratio is distinct from a running $\overline{\text{MS}}$ weak angle and from effective weak mixing angles extracted from neutral-current asymmetries. The manuscript therefore treats Eq. (13) as the primary target and later relates other schemes to it.

The following placement convention will be used. The Rivero–de Vries radical is read as a spectral object attached to the W/Z pole data. Other readings could attach the same radical to a high-scale boundary value, to a running $\overline{\text{MS}}$ coupling, or to an internal compactification

eigenvalue. Each reading changes the physical content of the proposal. The pole reading gives the word “spectrum” its literal electroweak meaning.

A vector-boson pole is defined by the complex pole of the propagator,

$$s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \quad (14)$$

Throughout this manuscript, $M_{V,\text{pole}}$ and $\Gamma_{V,\text{pole}}$ refer to the parameterization in Eq. (14). Martin and Robertson also record the alternative reporting convention

$$s_{\text{pole}} = (M - i\Gamma/2)^2. \quad (15)$$

Any input reported in that convention must be translated to Eq. (14) before Eq. (13) is formed [10]. Equivalently, s_V is the zero of the inverse transverse propagator after renormalization,

$$\Delta_{T,V}^{-1}(s_V) = 0. \quad (16)$$

When experimental inputs are quoted in a variable-width Breit–Wigner line-shape convention, the conversion to pole parameters must be performed before using Eq. (13). With a common W/Z variable-width convention, write

$$\gamma_V = \frac{\Gamma_{V,\text{BW}}}{M_{V,\text{BW}}}. \quad (17)$$

The pole parameters in Eq. (14) are then obtained as

$$M_{V,\text{pole}} = \frac{M_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}, \quad \Gamma_{V,\text{pole}} = \frac{\Gamma_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}. \quad (18)$$

Martin and Robertson summarize the corresponding vector-boson mass relation as

$$M_{V,\text{BW}}^2 = M_{V,\text{pole}}^2 + \Gamma_{V,\text{pole}}^2, \quad V = W, Z. \quad (19)$$

Equations (18)–(19) apply after the quoted W and Z inputs have been verified to share the same variable-width Breit–Wigner convention. Fixed-width line-shape inputs, already-pole inputs, and inputs using Eq. (15) require their stated convention map. This conversion is part of the source audit for the pole comparison [10].

The resulting convention chain is

$$(M_{V,\text{BW}}, \Gamma_{V,\text{BW}}) \longrightarrow (M_{V,\text{pole}}, \Gamma_{V,\text{pole}}) \longrightarrow s_V \longrightarrow s_{\text{pole}}^2. \quad (20)$$

A. Scheme map

The comparison uses the following scheme map before any weak-angle quantity is interpreted:

Object	Definition level	Role in this manuscript
s_{pole}^2	complex-pole mass ratio	primary low-energy spectral target
Breit–Wigner masses	experimental line-shape convention	input convention requiring Eq. (19)
$\hat{s}^2(Q)_{\overline{\text{MS}}}$	running coupling ratio in a renormalization scheme	scheme translation and loop-scale comparison
$\sin^2 \theta_{\text{eff}}^\ell$	neutral-current effective angle	phenomenological comparison after electroweak corrections
GUT boundary value	high-scale representation datum	comparison class for structural weak-angle values
compactification eigenvalue	internal spectral datum	possible origin for $\mathcal{O}_J$ after a dynamical derivation

The pole reading selects the first row. The remaining rows are translation targets. A future source-audited electroweak comparison should begin from the quoted experimental convention, map to complex pole quantities, form Eq. (13), and then compare with Eq. (9).

The inherited numerical illustration uses

$$M_W = M_Z \sqrt{1 - s_{\text{dv}}^2}. \quad (21)$$

The body of the paper defers a current numerical comparison until the quoted mass convention, complex-pole conversion, and uncertainty propagation are source-audited together. Appendix B records the inherited arithmetic for provenance. The CDF-II measurement is relevant only as an example showing that different line-shape inputs can move the descriptive pole-ratio comparison [11].

The Standard Model parameter literature distinguishes several vacuum and renormalization conventions. In particular, pure $\overline{\text{MS}}$ treatments may define the Higgs vacuum expectation value as the minimum of the loop-corrected effective potential in Landau gauge,

eliminating tadpole diagrams at the cost of gauge-choice dependence in intermediate parameters [10]. Such scheme details are relevant for comparing a mass-ratio statement with Lagrangian parameters, while Eq. (13) remains the pole-spectrum target.

Loop effects are therefore a conceptual discriminator. Corrections of order α/π are of the same kind of size one expects when moving between pole, effective, and running definitions. They ask what object the radical belongs to and at what scale it is to be read. A high-scale reading would make the DeVries value analogous to a GUT boundary datum. The pole-data reading makes it a low-energy spectral datum and transfers the burden to string/Kaluza-Klein dynamics.

B. Limited claim

The formal claim of this section is limited:

$$s_{\text{dV}}^2 \text{ is compared first with } 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (22)$$

Every comparison to $\overline{\text{MS}}$ inputs, effective angles, or line-shape masses requires an explicit map. Thus a disagreement among schemes diagnoses the placement of $\sqrt{J^2 + 4J}$ and the scale at which the radical is read.

IV. ELECTROWEAK EMBEDDING AS A PROJECTIVE DATUM

The gauge-Higgs sector gives the tree-level vector masses from the Higgs kinetic term

$$(D_\mu H)^\dagger (D^\mu H), \quad D_\mu = \partial_\mu - \frac{ig}{2} \sigma^a W_\mu^a - \frac{ig'}{2} B_\mu, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}. \quad (23)$$

The charged fields

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2) \quad (24)$$

receive a mass term $g^2 v^2 W_\mu^+ W^{-\mu}/4$. The neutral fields have mass matrix

$$\frac{v^2}{8} \begin{pmatrix} W_\mu^3 & B_\mu \end{pmatrix} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}. \quad (25)$$

It is diagonalized by

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \quad A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad \tan \theta_W = \frac{g'}{g}. \quad (26)$$

This gives

$$M_W^2 = \frac{g^2 v^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2 = 0. \quad (27)$$

The ratio is

$$1 - \frac{M_W^2}{M_Z^2} = \frac{g'^2}{g^2 + g'^2}. \quad (28)$$

Thus the vacuum scale v fixes the radial size of the massive vector pair, while the quotient fixes a projective direction in the two-coupling plane [5, 9].

A. Custodial structure and the electroweak ray

The Standard Model doublet can be packaged as the bidoublet

$$\Phi_H = \begin{pmatrix} H^{0*} & H^+ \\ -H^{+*} & H^0 \end{pmatrix}. \quad (29)$$

In the scalar sector this makes the approximate global $SU(2)_L \times SU(2)_R$ structure visible; the vacuum proportional to the identity preserves the diagonal custodial $SU(2)$ subgroup [9]. Hypercharge gauging and Yukawa splittings break the symmetry, while the tree-level doublet mass matrix preserves the relation

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1. \quad (30)$$

This is the electroweak reason the DeVries comparison should be a projective statement about the W/Z pair. The broken-to-unbroken deformation scales the vector masses with v while preserving the coupling-space direction.

The DeVries proposal is read in the same projective language:

$$\frac{M_W^2}{M_Z^2} \longleftrightarrow \frac{x_+(3/4)}{x_+(2)}. \quad (31)$$

The common scale is then supplied by the electroweak vacuum or, in a deeper model, by the compactification or brane scale that generates the effective gauge-Higgs sector.

The unbroken limit is the ray

$$v \rightarrow 0 \quad \text{with} \quad \frac{g'^2}{g^2 + g'^2} \text{ fixed}. \quad (32)$$

Along this ray, M_W and M_Z vanish together and the photon remains massless. This is the electroweak deformation relevant to the construction. The projective datum remains meaningful as the limiting direction even when the radial mass scale is taken to zero.

The central analytical question is the ordered assignment

$$(J_W, J_Z) = \left(\frac{3}{4}, 2\right). \quad (33)$$

The electroweak gauge fields are gauge-algebra fields before symmetry breaking: W^1, W^2, W^3 live in the adjoint of $SU(2)_L$, while B is the gauge field of the abelian $U(1)_Y$ factor. The value $J = 2$ has a direct $SU(2)_L$ adjoint reading through $T(T+1)$ with $T = 1$. The value $J = 3/4$ belongs to the Higgs/order-parameter side, where the symmetry-breaking field is an $SU(2)_L$ doublet with $T = 1/2$ and $T(T+1) = 3/4$.

This yields a candidate route:

$$J_Z = T_{\text{adj}}(T_{\text{adj}} + 1) = 2, \quad J_H = T_H(T_H + 1) = \frac{3}{4}. \quad (34)$$

The remaining calculation is the ordered sampling rule that turns the pair (J_H, J_{adj}) into the W/Z quotient in Eq. (31). A completed derivation must start from the gauge-Higgs Lagrangian, a current-algebra construction, or a compactification boundary problem and produce Eq. (33) from the same determinant that supplies the negative branch.

B. The assignment problem as a theorem target

The theorem target can be stated mechanism-independently. Let $\mathcal{E}$ be the electroweak gauge-Higgs system, including its projective W/Z pole data and scalar order-parameter channel. A DeVries derivation is a map

$$\mathcal{D} : \mathcal{E} \longrightarrow \mathcal{M}_J = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} \quad (35)$$

with the following outputs:

$$\mathcal{D}(\text{order-parameter channel}) \mapsto J_H = 3/4, \quad (36)$$

$$\mathcal{D}(\text{adjoint/current channel}) \mapsto J_{\text{adj}} = 2, \quad (37)$$

$$\frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}. \quad (38)$$

The theorem must also account for the negative branch in the same basis, so that the vector ratio and scalar clue arise from one operator.

A natural current to test is the Higgs isospin current

$$\mathcal{J}_\mu^a = iH^\dagger \frac{\sigma^a}{2} D_\mu H - i(D_\mu H)^\dagger \frac{\sigma^a}{2} H, \quad (39)$$

which transforms as an adjoint object while being built from the doublet order parameter. The current contains the two data needed for the ordered pair: doublet data enter the source, adjoint data enter the current. Turning this observation into Eq. (38) is the remaining electroweak theorem target.

V. NEGATIVE BRANCH AND THE SCALAR SIDE OF THE PROBLEM

The negative branch in Eq. (6) is fixed by the same secular equation as the positive branch. It is therefore part of the clue. Hans de Vries identified the positive-branch proximity to the mass-shell weak angle. The same two-root object also contains negative eigenvalues near the scalar/order-parameter scales of electroweak symmetry breaking when the positive branch is normalized by the vector spectrum. That observation is the main reason the numerics deserve conceptual attention.

The root identities imply

$$x_-(J) = -J - x_+(J) = -\frac{J}{x_+(J)}. \quad (40)$$

Any physical use of the positive branch automatically carries a partner branch.

The Higgs potential can be written as

$$V(H) = m^2 H^\dagger H + \lambda (H^\dagger H)^2, \quad m^2 < 0 \quad (41)$$

in the broken phase. The sign of the quadratic term makes the negative branch tempting as an order-parameter datum. If the normalization is fixed by $M_Z^2 = \mu^2 x_+(2)$, the two negative magnitudes are

$$M_{-,1} = M_Z \sqrt{\frac{|x_-(2)|}{x_+(2)}}, \quad M_{-,1/2} = M_Z \sqrt{\frac{|x_-(3/4)|}{x_+(2)}}. \quad (42)$$

Under this normalization, the two branch magnitudes fall in the range of familiar electroweak scalar scales. They are unresolved branch-scale identifications. The scalar reading becomes a physical statement only after the same determinant, normalization, and scheme are derived from the gauge-Higgs or source-theory data.

A valid identification must be gauge-invariant and scheme-controlled. Possible targets include m^2 , λv^2 , the Higgs pole mass, or an eigenvalue in a compactification problem. The branch values associated with $J = 3/4$ and $J = 2$ generate definite dimensionless numbers. The map from those numbers to scalar observables remains open.

A. Logical role of the negative root

The negative root changes the character of the clue. The full quadratic gives a paired object: a pole-ratio candidate and a scalar-side candidate arise from one secular equation. This pairing supplies the main conceptual reason to retain the negative branch. A dynamical theory that produces Eq. (2) has to say what both branches mean in the same basis.

The two-root structure can be stated as a spectral chain,

$$\mathcal{M}_J \longrightarrow \begin{cases} x_+(J) & \text{projective vector datum,} \\ x_-(J) & \text{scalar/order-parameter datum.} \end{cases} \quad (43)$$

The first line is the pole-ratio reading developed in Sec. III. The second line is a target for the Higgs, boundary, brane, or compactification sector. The chain is meaningful because both entries come from the same diagonalization. Any proposed derivation that keeps the positive branch acquires the negative-branch obligation automatically.

This obligation also sharpens the electroweak assignment. The ordered pair $(J_H, J_{\text{adj}}) = (3/4, 2)$ already suggests two electroweak structures: the doublet order parameter and the adjoint current. The negative branch asks whether the doublet/order-parameter side can be read as a scalar instability, a vacuum scale, or a deformation mode in the same operator that gives the vector pole quotient. The scalar reading therefore ties together three open calculations: the ordered J -assignment, the determinant derivation, and the branch-to-Higgs map.

B. Gauge-invariant scalar language

The scalar sector has several layers. The Higgs doublet H itself is gauge-covariant. The radial composite

$$\varphi^2 = 2H^\dagger H \quad (44)$$

is the natural gauge-invariant local order-parameter variable for this manuscript's purpose. In the broken phase, a gauge choice may represent the vacuum by a real radial expectation value, yet the physical statement should be expressible through invariant quantities, pole positions, or matched parameters. The negative branch therefore requires a defined scalar functional before it can carry a physical claim.

An abstract target is

$$\mathcal{F}_{\text{sc}} = \mathcal{F}_{\text{sc}}(H^\dagger H, (D_\mu H)^\dagger D^\mu H, V(H), \text{boundary or compactification data}), \quad (45)$$

with a branch assignment

$$\mathcal{F}_{\text{sc}}(J_\star) = C_{\text{sc}} |x_-(J_\star)|. \quad (46)$$

Here $J_\star$, C_{sc} , and the scheme are part of the theorem. The expression may describe a pole mass, a curvature of the scalar potential, a vacuum-ray coordinate, a brane separation, or a compactification deformation. The manuscript should treat all of these as candidate targets until one route derives a specific object.

The scalar pole target is the most restrictive because it demands the same pole discipline as the vector comparison. The potential target is more sensitive to matching conventions, because m^2 , λ , and v depend on the renormalization scheme and on how the effective potential is defined. The compactification target is broader: $x_-(J)$ may first appear as an eigenvalue or deformation parameter in a higher-dimensional problem, with the four-dimensional Higgs interpretation emerging after reduction.

C. Vacuum ray and branch pairing

The electroweak ray separates scale from direction. Along the broken-to-unbroken ray, the radial coordinate scales the vector masses, while the ratio

$$\frac{M_W^2}{M_Z^2} = \frac{g^2}{g^2 + g'^2} \quad (47)$$

is projective at tree level. The DeVries quotient addresses this projective datum. The negative branch is naturally assigned to the scalar side because it shares the same two-channel operator while carrying the opposite eigenvalue.

In a route with fields h_J and a_J , the branch vectors have the schematic form

$$u_\pm(J) = \alpha_\pm(J)h_J + \beta_\pm(J)a_J. \quad (48)$$

The scalar-side reading becomes a statement about the h_J component of $u_-(J)$. The vector-side reading becomes a statement about the projective pole ratio extracted from $u_+(J)$. A complete derivation should therefore give a branch basis, an inner product, and a normalization. These data determine whether the negative eigenvector genuinely selects an order-parameter amplitude or whether it belongs to an auxiliary sector of the chosen source theory.

The electroweak ray also supplies a consistency check. Taking the radial vacuum coordinate toward the symmetric point should collapse the vector masses together with the Higgs order parameter. The DeVries ratio can remain as a projective direction in coupling or spectral space. A scalar interpretation of x_- should respect this ray structure by scaling with the radial data or by controlling the deformation that sets the radial scale.

D. Candidate scalar targets

The Standard Model scalar sector gives several inequivalent objects that can carry a scalar-side interpretation [8, 9]. They have different scheme and gauge properties:

Target	Meaning	DeVries requirement
m^2	quadratic term in the Higgs potential	define the scheme and gauge-invariant extraction
λv^2	curvature scale of the scalar potential along the radial mode	connect branch value to the electroweak ray
$M_{h,\text{pole}}^2$	scalar pole position	derive the scalar pole from the same determinant as W/Z
radion/modulus value	eigen-compactification scalar or boundary deformation	identify the compact operator and its normalization
brane separation mode	endpoint or brane order parameter	show coupling to the adjoint/current channel

The pole interpretation is the sharpest target. It asks the same object that governs the vector pole quotient to generate a scalar pole or scalar instability. The potential interpretation is more flexible, because m^2 , λ , and v depend on conventions and matching. The compactification interpretation is broader still; it may identify the negative branch with

a modulus or boundary deformation whose four-dimensional scalar meaning emerges after reduction.

E. Route-specific scalar maps

The endpoint route maps the negative branch to brane scalar data. In a D-brane system, transverse positions of branes appear as scalar fields on the worldvolume, while strings stretched between branes acquire masses controlled by separation data [2, 3]. A DeVries endpoint derivation would identify $u_-(J)$ with a brane-Higgsing direction, a stretched-string scalar, or a boundary order-parameter field. The map has to use the same endpoint algebra that supplies the current trace and off-diagonal product.

The interval route maps the negative branch to boundary or modulus data. A five-dimensional gauge field supplies four-dimensional vector modes and scalar components; the compactification radius, brane terms, and boundary conditions determine the light spectrum [7, 12]. A DeVries interval derivation would identify $u_-(J)$ with a boundary scalar amplitude, an A_5 -like scalar, a radion-like deformation, or a brane-localized order parameter. The map has to preserve the same boundary determinant that gives the W/Z branch quotient.

The G_2 route maps the negative branch to singularity deformation data. Local G_2 constructions place gauge sectors and charged matter at singular loci, and Higgs-bundle descriptions encode matter localization and interaction data through functions on an associative three-cycle [14–16]. A DeVries G_2 derivation would identify $u_-(J)$ with a deformation parameter, a localized modulus, a critical value of f_Q , or a charge-density functional. The same local geometry then has to produce the adjoint/current and order-parameter entries of the kernel.

These maps give the negative branch a practical role in route selection. A candidate route that produces the positive-branch quotient and leaves the negative eigenvector with a scalar or deformation interpretation gains additional physical content. A route that assigns the negative branch to an auxiliary variable defines a narrower physical claim.

F. Eigenvectors and branch meaning

The two roots also have eigenvectors. For the matrix in Eq. (4), an eigenvector with eigenvalue x can be represented by

$$v_x \propto \begin{pmatrix} \sqrt{J} \\ x \end{pmatrix}. \quad (49)$$

Thus the branch assignment is a statement about channel composition. The positive and negative roots are mixtures of the same two basis amplitudes. A scalar interpretation of x_- should therefore identify the basis in which one component is the vector/current side and the other is the order-parameter or boundary side.

A criterion is the following. If the same two-channel determinant produces both the vector pole ratio and the scalar instability, then the determinant should appear in an effective action, boundary condition, or self-energy equation. The corresponding theorem target is an explicit determinant with an identified second-channel field or mode.

VI. STRING, KALUZA–KLEIN, AND G_2 COMPLETION PROBLEM

The secular equation has the form of a low-level branch splitting that preserves a high-spin datum. This is the reason string and Regge language enter the construction. The original Rivero construction imposed a high-spin condition motivated by Regge behavior [1]. A string interpretation must therefore preserve the asymptotic slope while explaining why the low levels split through the two-channel determinant.

The string-theory input is concrete. Quantized strings generate towers, level matching, endpoint sectors, winding, and compactification charges. A compact circle already contains the basic Kaluza–Klein mechanism: a higher-dimensional metric reduces to a lower-dimensional metric, a gauge field, and a radius modulus; Fourier modes along the circle acquire masses n^2/R^2 and charges under the gauge field descending from the metric [2]. The string worldsheet adds winding sectors, and winding modes couple to the vector descending from the antisymmetric tensor. Compactification therefore supplies more than one integer or representation label, and different sectors can couple to different gauge fields.

Open strings add boundary data. Neumann and Dirichlet boundary conditions relate left- and right-moving oscillators differently, and D-branes localize open-string states on subman-

ifolds [2]. Coincident branes generate nonabelian gauge fields from open-string excitations. A DeVries derivation based on endpoints or branes must identify the two channels in Eq. (50) with actual boundary degrees of freedom, endpoint charges, or brane/bulk mixing modes.

The older phenomenological string lineage is relevant here. Regge trajectories in hadron spectra motivate a rotating flux-tube picture, and the open-string Veneziano amplitude was first written to capture strong-interaction features before QCD supplied the gauge-theory description [2]. In this lineage, a low-energy spectral regularity can guide the search for a deeper string or boundary structure, provided the proposed determinant is eventually derived.

Endpoint physics gives the simplest analogy for charged vector states. In a coincident-brane system, strings stretching between branes carry off-diagonal gauge quantum numbers, and the massless excitations organize into a nonabelian gauge field [2]. The electroweak target is a broken nonabelian gauge sector coupled to a doublet order parameter. A successful endpoint model must therefore produce both the adjoint-current datum $J = 2$ and the doublet/order-parameter datum $J = 3/4$ inside one boundary or brane determinant.

A Kaluza–Klein or brane derivation would have to produce the determinant

$$\det \begin{pmatrix} x & -\sqrt{J} \\ -\sqrt{J} & x + J \end{pmatrix} = 0. \quad (50)$$

This can arise from a two-channel boundary problem, a mixing of brane-localized and bulk modes, or an effective orbit problem. The required output is the exact J -dependence in Eq. (50).

Extra-dimensional electroweak models show that boundary conditions can break gauge symmetry and determine vector spectra. Higgsless constructions on intervals provide a controlled language for relating boundary data, vector masses, fermion localization, and precision observables [7]. The interval viewpoint is valuable because boundary conditions can be obtained as limits of boundary dynamics. For the present construction, this gives a precise target: derive the $J = 3/4, 2$ samples as boundary data associated with the Higgs doublet/order-parameter channel and the adjoint/current channel.

Witten’s search for realistic Kaluza–Klein theory gives the historical benchmark. Extra dimensions can generate gauge groups from geometry, including $SU(3) \times SU(2) \times U(1)$, while the realistic fermion quantum numbers impose severe constraints [12]. The DeVries

construction inherits this lesson. A compactification mechanism must solve the electroweak spectral problem and remain compatible with chiral matter.

M-theory on G_2 -holonomy manifolds supplies a separate completion map. Smooth compact G_2 compactifications give four-dimensional minimal supersymmetry in the large smooth limit. Realistic charged chiral matter requires singularities supporting nonabelian gauge sectors and chiral fermions [13–15]. Local Higgs-bundle descriptions over associative three-cycles give a modern way to organize gauge sectors, localized matter, and singular transitions [16].

The completion problem can therefore be stated as a list of mechanisms and obligations.

Mechanism	What it can supply	Required DeVries test
Regge/string tower	high-spin asymptotics	preserve slope and produce Eq. (50)
Circle KK reduction	gauge fields and charged KK modes	identify the charge/Casimir source of J
Winding sector	second compactification charge	test a winding-like interpretation of the second branch
Interval/brane BC	vector spectra and boundary dynamics	derive $J = 3/4, 2$ samples from boundary data
Endpoint/open-string sector	localized gauge states	identify the two channels as endpoint or brane/bulk modes
Singular G_2	gauge/chiral sectors	localize the electroweak operator geometrically
Higgs bundle	matter/mass localization	identify the two branches as spectral-cover data

VII. RADICAL PLACEMENT AND SCALE ATTACHMENT

The secular equation contains the radical

$$\Delta_J^{1/2} = \sqrt{J^2 + 4J}. \quad (51)$$

The manuscript treats $\Delta_J^{1/2}$ as the central object whose physical address must be identified. The algebra alone specifies a two-branch spectrum. Physics begins when the radical is

assigned to a pole condition, a boundary determinant, a compactification eigenvalue, a worldsheet constraint, or a running matching condition.

A. Placement options

The possible placements are distinct research programs.

Placement	Physical meaning	Required derivation
Pole spectrum	$\Delta_J^{1/2}$ enters a propagator or self-energy pole condition	derive the W/Z pole ratio and scalar partner branch from one spectral equation
GUT boundary	$\Delta_J^{1/2}$ fixes a high-scale weak-angle datum	embed the J -samples in a unification representation and run to the pole scale
Running parameter	$\Delta_J^{1/2}$ constrains a scheme-dependent coupling or vacuum parameter	specify the scheme, scale, matching map, and threshold corrections
KK eigenvalue	$\Delta_J^{1/2}$ comes from a compact internal wave operator	identify the internal operator, mode labels, degeneracies, and boundary conditions
Open-string endpoint	$\Delta_J^{1/2}$ comes from or Chan–Paton data	derive the two-channel matrix from endpoint charges and brane/bulk mixing
Worldsheet constraint	$\Delta_J^{1/2}$ comes from matching, conformal weights, or a sector	identify the CFT labels and show how they reduce to the electroweak determinant

The pole-spectrum placement is the current working reading. It gives the numerical clue its electroweak content because W and Z are observed as poles. It also gives the negative branch a task: the same determinant should know about the scalar/order-parameter side. This reading demands the most direct dynamical calculation, since it asks string/Kaluza–Klein structure to survive down to a low-energy pole observable.

The GUT-boundary placement gives a familiar comparison class. In SU(5) and Spin(10) settings, representation theory can give a clean weak-angle value at the high scale, with running and thresholds supplying the low-energy relation [20]. A DeVries high-scale reading

would join that lineage. The current manuscript pursues the pole reading because the negative branch and pole-ratio clue appear together in the same low-energy normalization.

B. Scale attachment

Let $\mathcal{O}_J$ denote the physical operator whose two-channel reduction gives

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J. \quad (52)$$

The scale question is then a question about where $\mathcal{O}_J$ lives:

$$\mathcal{O}_J \in \{\mathcal{O}_{\text{pole}}, \mathcal{O}_{\text{GUT}}, \mathcal{O}_{\text{KK}}, \mathcal{O}_{\text{brane}}, \mathcal{O}_{\text{CFT}}\}. \quad (53)$$

Each address has a different interpretation of the same algebra. A pole operator must be built from four-dimensional gauge-Higgs correlators. A GUT operator must be built from representation traces and matching. A KK operator must be a compact internal differential operator or boundary value problem. A brane operator must couple localized and bulk modes. A worldsheet operator must arise from conformal weights, current algebra, or level matching.

This list fixes the scheme dependence of the radical. A different placement changes the required derivation.

C. Placement chain

The placement question can be recorded as a chain. A source theory $\mathcal{S}$ supplies an operator, the operator supplies a two-channel kernel, the kernel supplies branches, and the branches acquire physical meaning after a scheme and an assignment rule have been fixed:

$$\begin{aligned} \mathcal{S} &\longrightarrow \mathcal{O}_J \longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\}, \\ \{x_+(J), x_-(J)\} &\longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}\}. \end{aligned} \quad (54)$$

Here $\mathcal{F}_{\text{sc}}$ denotes the scalar/order-parameter functional sought in Sec. V, and $\mathcal{C}_{\text{top}}$ denotes the global-form and representation constraints discussed in Secs. XIV and XV. The chain

separates four tasks:

$$\text{operator origin: } \mathcal{S} \mapsto \mathcal{O}_J, \quad (55)$$

$$\text{two-channel reduction: } \mathcal{O}_J \mapsto \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad (56)$$

$$\text{branch reading: } \mathcal{M}_J \mapsto \{x_+(J), x_-(J)\}, \quad (57)$$

$$\text{physical assignment: } \{x_{\pm}\} \mapsto (s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}). \quad (58)$$

Each arrow has different source support. Complex-pole and scheme language belongs to the last arrow [10]. Higgs doublet, adjoint current, custodial structure, and the electroweak ray belong to the assignment arrow [8, 9]. Endpoint labels, D-brane gauge fields, and interval boundary conditions belong to the first two arrows [2, 3, 7]. Equation (57) is algebraically controlled; the other arrows are open derivation targets.

This chain also organizes the negative branch. If a candidate source theory derives $\mathcal{M}_J$, then $x_-(J)$ is carried by the same light-sector basis as $x_+(J)$. The scalar question becomes a question about the eigenvector map,

$$u_-(J) \in \mathcal{H}_J \quad \longmapsto \quad \mathcal{F}_{\text{sc}}, \quad (59)$$

where $\mathcal{H}_J$ is the two-dimensional light sector. A Higgs interpretation requires $\mathcal{F}_{\text{sc}}$ to be gauge invariant. A brane interpretation requires $\mathcal{F}_{\text{sc}}$ to be a modulus, separation, or localized boundary datum. A compactification interpretation requires $\mathcal{F}_{\text{sc}}$ to be a normalized internal eigenmode or deformation parameter.

The chain gives a compact criterion for acceptance. A proposed mechanism should fill one of the arrows in Eq. (54) or state a new arrow with its source theory, scheme, and physical assignment.

D. Original research route

The route developed here is:

$$\begin{aligned} \text{Regge/high-spin datum} &\longrightarrow \text{two-channel secular determinant} \\ &\longrightarrow \text{electroweak pole ratio and scalar partner branch.} \end{aligned} \quad (60)$$

The first arrow belongs to the string/dual-model tradition: spectra organize into towers with controlled asymptotics. The second arrow is the DeVries algebra. The third arrow is the open electroweak calculation.

A concrete model should therefore supply four ingredients:

1. an internal label whose quadratic invariant is J ;
2. a two-channel system whose off-diagonal mixing is fixed by $\sqrt{J}$;
3. a diagonal trace term giving the $-J$ entry;
4. a physical normalization that places the positive branch on the vector pole ratio and the negative branch on the scalar/order-parameter side.

The fourth ingredient is the new conceptual burden. It turns the algebraic clue into a physical program and fixes the relation between the pole reading and high-scale weak-angle boundary data.

VIII. DUAL-MODEL LINEAGE AND OPEN-STRING ENDPOINT DATA

The DeVries construction enters the manuscript through a spectral regularity. This is why the older dual-model lineage matters. The Veneziano amplitude was introduced to capture strong-interaction regularities before its string interpretation became clear; its open-string form later supplied a world-sheet explanation of the pole tower and the associated Regge behavior [2]. The lesson for the present work is methodological and technical. A compact algebraic regularity in particle data can guide a deeper spectral construction when the amplitude or boundary problem subsequently explains the regularity.

The relevant feature is the coexistence of a low-level spectrum and a high-spin asymptotic rule. In string amplitudes, pole residues organize the spins carried by the exchanged string states, and the tower structure follows from the quantized oscillators [2]. In the DeVries program, the secular equation is required to do something analogous at a much lower and more constrained level:

$$\text{high-spin/Regge regularity} \quad \rightsquigarrow \quad x^2 + Jx - J = 0 \quad \rightsquigarrow \quad \{x_+(J), x_-(J)\}. \quad (61)$$

The first arrow is historical motivation until a world-sheet or boundary calculation produces the second expression. The second arrow is algebraic and already fixed. The manuscript therefore treats the Rivero/DeVries observation as a clue that selects a target determinant.

Open strings provide a more concrete mechanism than closed-string Regge language alone. An open-string state carries oscillator data in the interior and boundary data at its endpoints. D-branes give a geometric home to the endpoints, and coincident branes give nonabelian gauge fields from open-string modes [2, 3]. This suggests a precise kind of model for Eq. (50): one channel should be an adjoint or current channel associated with gauge fields, while the other should carry the order-parameter data that samples the doublet invariant.

For the DeVries determinant, the endpoint route can be written as a quadratic action for two boundary-coupled amplitudes,

$$S_J^{(2)} = \begin{pmatrix} \psi_1^\dagger & \psi_2^\dagger \end{pmatrix} \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \quad (62)$$

with the DeVries target

$$\kappa_J^2 = J, \quad \tau_J = J. \quad (63)$$

Here ψ_1, ψ_2 are placeholders for actual endpoint, brane, or localized modes. The notation is intentionally sparse. The physics is carried by the derivation of κ_J and τ_J .

There are three natural endpoint readings.

1. *Chan-Paton/current reading.* The endpoint labels carry gauge indices. The diagonal entry τ_J is a trace or current-algebra invariant, and the off-diagonal entry κ_J is a group-theoretic transition matrix element.
2. *Brane/bulk reading.* One amplitude is localized on a brane or boundary, while the other propagates in the bulk. The determinant comes from integrating over the bulk mode and imposing the boundary matching condition.
3. *Order-parameter reading.* One channel is an adjoint-current datum and the other is a doublet/order-parameter datum. This is the route that can explain why the electroweak samples are $J = 2$ and $J = 3/4$.

Each reading must respect the same electroweak constraint. The pole ratio compares charged and neutral vector poles, while the $J = 3/4$ datum belongs naturally to the Higgs

doublet side. A successful endpoint construction therefore has to turn an order-parameter doublet invariant into the charged-vector sample in the pole quotient. This is the ordered-sampling problem in boundary language.

The open-string route also clarifies the negative branch. The two roots are eigenvalues of one two-channel system. If the positive branch is associated with the vector pole quotient, the negative branch should correspond to the partner channel in the same boundary problem. In an endpoint construction this partner may be a scalar boundary modulus, a Higgs-like order parameter, a brane separation mode, or an auxiliary channel eliminated in the low-energy action. The manuscript uses the negative branch as a demand on the model: the same determinant has to explain both eigenvalues.

This places the string interpretation in a concrete form. The target is an endpoint or brane calculation whose reduced determinant has the entries in Eq. (63), whose low-energy normalization gives the W/Z pole quotient, and whose partner eigenvalue lands in the scalar/order-parameter sector.

IX. KALUZA–KLEIN SPECTRA, BOUNDARY CONDITIONS, AND SECULAR DETERMINANTS

Kaluza–Klein theory supplies the second concrete language for the DeVries program. The original idea unifies gauge fields with geometry by expanding a higher-dimensional theory around a compact ground state. In Witten’s account, symmetries of the compact space appear as gauge symmetries in the lower-dimensional theory, and the physical spectrum is obtained from small oscillations around the compactification background [12]. The same account also records the hard model-building burden: obtaining the Standard Model gauge group, fermion quantum numbers, and scalar sector is a serious representation problem, even when the gauge group itself can be engineered.

The DeVries problem asks for a narrower and sharper object inside this tradition: a compact or interval derivation of the two-channel determinant

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (64)$$

This determinant is the entire target of the KK route.

A. Circle compactification and two charge channels

String compactification on a circle gives a clean source-backed example of a two-channel spectral structure. Ordinary Kaluza–Klein momentum is quantized along the circle. Closed strings add winding around the same circle. Polchinski’s circle example shows that the Kaluza–Klein vector couples to compact momentum, while the vector descending from the antisymmetric tensor couples to winding [3]. At the self-dual radius the spectrum has enhanced current algebra, and changing the radius is described by an operator built from the same currents.

This gives a source-backed analogy:

$$\left(\begin{array}{c} \text{momentum-like channel} \\ \text{winding-like or boundary channel} \end{array} \right) \rightsquigarrow \{x_+(J), x_-(J)\}. \quad (65)$$

The analogy becomes a derivation after the two channels are reduced to Eq. (64). The self-dual-radius example also explains why a scalar modulus belongs in the discussion. The compact radius is a scalar field in the lower-dimensional theory, and gauge enhancement can occur at special values of that scalar. This is the compactification version of the manuscript’s positive/negative branch problem: a vector spectral statement and a scalar modulus can arise from the same compact sector.

B. Interval variational data

Extra-dimensional field theory on an interval gives a more direct determinant language. Csaki–Hubisz–Meade derive boundary conditions from the variational principle for fields on an interval and show how boundary mass terms and boundary kinetic terms modify those conditions [7]. A scalar example has the schematic action

$$S_{\text{int}} = \int d^4x \int_0^L dy \frac{1}{2} \left[\partial_M \Phi \partial^M \Phi - V(\Phi) \right] + S_0[\Phi] + S_L[\Phi], \quad (66)$$

with boundary functionals S_0 and S_L . The boundary variation supplies equations at $y = 0$ and $y = L$. Boundary mass terms give Robin-type conditions. Boundary kinetic terms can make the boundary equation depend on the four-dimensional eigenvalue. This is exactly the structural feature needed by a secular determinant.

For a mode $\Phi(x, y) = \phi_\lambda(x)f_\lambda(y)$, the interval problem has the abstract form

$$\mathcal{L}_J f_\lambda(y) = \lambda f_\lambda(y), \quad 0 < y < L, \quad (67)$$

$$\mathcal{B}_{0,J}(\lambda)f_\lambda(0) = 0, \quad \mathcal{B}_{L,J}(\lambda)f_\lambda(L) = 0. \quad (68)$$

The DeVries route asks for a two-dimensional light sector $\mathcal{H}_J$ extracted from this problem.

A convenient basis is

$$\mathcal{H}_J = \text{span}\{h_J, a_J\}, \quad (69)$$

where h_J is a boundary order-parameter amplitude and a_J is an adjoint/current amplitude.

The interval derivation then reduces the bulk-plus-boundary problem to an effective kernel

$$K_J^{\text{int}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{\text{int}}(\lambda) & \Sigma_{ha,J}^{\text{int}}(\lambda) \\ \Sigma_{ah,J}^{\text{int}}(\lambda) & \lambda + \Sigma_{aa,J}^{\text{int}}(\lambda) \end{pmatrix}. \quad (70)$$

The DeVries target is the boundary limit

$$\Sigma_{hh,J}^{\text{int}} = 0, \quad \Sigma_{aa,J}^{\text{int}} = J, \quad \Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J. \quad (71)$$

This is the interval version of the endpoint target in Sec. XI.

C. Dimensional Schur-complement target

The dimensional interpolation of Appendix D selects a concrete interval calculation. Let the middle-dimensional sector be split into a light boundary subspace $\mathcal{H}_J = \text{span}\{h_J, a_J\}$ and a tower, bulk, or compact sector with operator L_J . With a boundary kernel $K_J^{\text{bdry}}(\lambda)$, coupling map V_J , and specified inner product, integrating out the heavy sector gives

$$K_J^{\text{eff}}(\lambda) = K_J^{\text{bdry}}(\lambda) - V_J^\dagger(\lambda - L_J)^{-1}V_J. \quad (72)$$

The active calculation target is

$$K_J^{\text{eff}}(\lambda) \longrightarrow \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} \quad (73)$$

in a limit where additional light channels decouple or acquire a controlled block-diagonal form. This formulation connects the $D = 10/6$ middle line to the route kernel: K_J^{bdry} contains boundary kinetic and mass data, L_J contains the integrated tower, and V_J contains the order-parameter/current mixing. A successful interval derivation must specify these three objects, the inner product, and the normalization yielding Eq. (71).

D. Dictionary for the interval entries

The interval construction separates the entries of Eq. (64) into physically addressable data:

Entry	Interval origin	Derivation target
$\Sigma_{hh,J}^{\text{int}} = 0$	order-parameter boundary	reference fix the subtraction by a symmetry, a brane condition, or a normalization convention
$\Sigma_{aa,J}^{\text{int}} = J$	boundary inertia, adjoint current trace, or brane kinetic term	identify the quadratic invariant that appears in the boundary equation
$\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$	overlap between boundary and bulk profiles, brane-localized normalized overlap or gener-mixing, or endpoint-current coupling	derive the invariant from a ator product
$(J_H, J_{\text{adj}}) = (3/4, 2)$	Higgs/order-parameter channel and adjoint/current channel	derive the ordered electroweak sampling rule from the coupled interval problem

The table is intentionally strict. A boundary mass term by itself can produce many secular equations. The DeVries claim requires the trace and determinant data in Eq. (71) and the electroweak ordering of J -samples.

The possible origins of J are correspondingly limited:

$$J \in \{C_2(R_{\text{int}}), C_2(R_{\text{bdry}}), h_{\text{CFT}}, q_{\text{KK}}^2, \langle f_{\text{bdry}}, f_{\text{bulk}} \rangle^2\}. \quad (74)$$

Here C_2 denotes a quadratic invariant, h_{CFT} a conformal weight or current-algebra datum, q_{KK} a compact charge, and the overlap term a normalized boundary/bulk integral. The final paper should name one of these entries and derive the equality to J .

E. Gauge fields, boundary breaking, and the electroweak ray

Gauge fields on an interval carry a stronger constraint than scalar toy models. Boundary conditions that break gauge symmetry have to arise from the action and preserve the con-

sistency of the lower-dimensional gauge theory. Csaki–Hubisz–Meade use this language for electroweak symmetry breaking by boundary conditions and for Higgsless models [7]. Their framework is relevant here because it gives a controlled setting in which massive vector modes, a photon zero mode, custodial structure, and brane-localized terms can be discussed in one spectral problem.

The DeVries construction requires the boundary model to preserve the electroweak ray. The low-energy gauge-Higgs mass matrix has the form

$$\mathcal{M}_{\text{EW}}^2(v) = v^2 \mathcal{P}(g, g'), \quad (75)$$

where v is radial and $\mathcal{P}(g, g')$ is projective. The DeVries ratio can act on $\mathcal{P}(g, g')$, or on a boundary operator whose pole condition reduces to that same projective datum. Thus the interval model should yield a coupled charged/neutral construction:

$$\left[\begin{array}{c} SU(2)_L \text{ charged sector} \\ SU(2)_L \times U(1)_Y \text{ neutral sector} \\ \text{photon zero mode} \end{array} \right] \longrightarrow \frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}. \quad (76)$$

This requirement places the W and Z readings inside one gauge-Higgs or boundary system. The photon must remain the unbroken mode, and the massive neutral pole must carry the adjoint/current sample.

The interval version of the ordered assignment theorem is:

$$\begin{aligned} &\text{bulk gauge fields} + \text{boundary terms} + \text{order-parameter channel} \\ &\implies K_J^{\text{int}}(\lambda) \implies (J_W, J_Z) = (3/4, 2). \end{aligned} \quad (77)$$

This theorem is parallel to the endpoint theorem in Sec. XI. The two routes can eventually compete: endpoint labels emphasize matrix algebra, while interval boundary conditions emphasize variational origin and pole equations.

F. Scalar partner from interval data

The negative branch also has a KK address. Witten’s review notes that compactification naturally produces scalar degrees of freedom, including radius moduli and possible charged scalar modes tied to symmetry breaking questions [12]. Polchinski’s circle compactification likewise identifies the compact radius with a lower-dimensional scalar and shows that

changing the radius is tied to current algebra at enhanced-symmetry points [3]. In interval language, scalar data can appear as boundary values, Wilson-line-like data, brane-localized order parameters, or moduli of the compact geometry.

The scalar-branch target is therefore

$$u_-(J) \in \mathcal{H}_J \quad \longmapsto \quad \mathcal{F}_{\text{KK}} \in \{R(x), H^\dagger H, A_5, \text{brane scalar, localized modulus}\}. \quad (78)$$

Here $u_-(J)$ is the negative-branch eigenvector of the two-channel kernel. A final derivation should identify $\mathcal{F}_{\text{KK}}$, state its gauge transformation properties, and explain its scheme relation to the low-energy Higgs/order-parameter variables.

G. KK completion criterion

The KK route is fixed by the following completion chain:

$$\begin{aligned} \text{compact or interval source} &\longrightarrow \text{boundary variational problem} \longrightarrow K_J^{\text{int}}(\lambda), \\ K_J^{\text{int}}(\lambda) &\longrightarrow \lambda^2 + J\lambda - J \longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{KK}}\}. \end{aligned} \quad (79)$$

The first arrow is source-backed by the KK and interval literature. The second arrow is the DeVries determinant derivation. The last arrow is the pole-placement and scalar-branch assignment. This chain gives the manuscript a concrete way to compare compactification, endpoint, and G_2 routes.

X. TOY ENDPOINT-BOUNDARY OPERATOR

The previous sections list possible mechanisms. This section turns those mechanisms into a minimal toy operator. The toy model is a derivation target with named matrix entries.

Let $h_J(x)$ denote a boundary-localized order-parameter amplitude and $a_J(x)$ denote an adjoint/current amplitude. The quadratic light-sector action is

$$S_J^{(2)} = \int d^4x \begin{pmatrix} h_J^\dagger & a_J^\dagger \end{pmatrix} \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} h_J \\ a_J \end{pmatrix}. \quad (80)$$

Here λ is the spectral variable after normalization. The DeVries target is

$$\tau_J = J, \quad \kappa_J^2 = J, \quad (81)$$

which gives

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (82)$$

The fields h_J and a_J are placeholders with fixed jobs. The h_J channel must carry the doublet/order-parameter invariant when $J = 3/4$. The a_J channel must carry the adjoint/current invariant when $J = 2$. The off-diagonal entry κ_J is the physical mixing amplitude between the two channels. The diagonal entry τ_J is the trace datum of the adjoint/current side.

A. Boundary origin

Interval field theory supplies a mechanism for such a matrix. Boundary mass and kinetic terms modify the variational boundary conditions, and brane-localized kinetic terms can make the boundary condition depend on the four-dimensional eigenvalue [7]. A schematic bulk-plus-boundary system is

$$S_{\text{bulk}} = \int d^4x \int_0^L dy \frac{1}{2} \left[(\partial_M A_J)^2 + (\partial_M H_J)^2 \right] + S_{\text{bdry}}[A_J, H_J]. \quad (83)$$

After solving the bulk equations and restricting to the two light boundary amplitudes, the effective inverse propagator has the form

$$K_J^{\text{eff}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}(\lambda) & \Sigma_{ha,J}(\lambda) \\ \Sigma_{ah,J}(\lambda) & \lambda + \Sigma_{aa,J}(\lambda) \end{pmatrix}. \quad (84)$$

The DeVries calculation asks for a normalization and a boundary limit in which

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J. \quad (85)$$

This formulation gives each matrix entry a physical address. $\Sigma_{hh,J}$ tests whether the order-parameter channel can be used as the zero reference. $\Sigma_{aa,J}$ tests whether the adjoint/current channel supplies the trace J . $\Sigma_{ha,J}$ tests whether endpoint, brane, or overlap physics supplies $\sqrt{J}$.

B. Endpoint interpretation

In an open-string or D-brane reading, a_J is an endpoint current or Chan–Paton-like gauge excitation, while h_J is a boundary order-parameter mode. Coincident branes naturally give

nonabelian gauge fields from endpoint sectors [2, 3]. The toy model demands a more specific result: the endpoint algebra must reduce to the entries in Eq. (81).

The corresponding route is

$$\begin{aligned}
\text{endpoint labels} &\longrightarrow \kappa_J^2 = J \\
\text{adjoint/current trace} &\longrightarrow \tau_J = J \\
\text{boundary normalization} &\longrightarrow \Sigma_{hh,J} = 0.
\end{aligned} \tag{86}$$

This table defines the calculation. It also gives a compact way to compare candidate completions.

C. Electroweak matching

The toy operator must match the electroweak projective data. The desired chain is

$$\begin{aligned}
\text{boundary data} &\longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\} \\
&\qquad\qquad\qquad \downarrow \\
&\qquad\qquad\qquad \text{W/Z pole ratio and scalar branch.}
\end{aligned} \tag{87}$$

The positive branch should yield

$$\frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}, \quad (J_H, J_{\text{adj}}) = (3/4, 2). \tag{88}$$

The negative branch should identify the scalar or boundary partner in the same basis. The toy model therefore joins the three open calculations: ordered electroweak sampling, determinant derivation, and scalar-branch interpretation.

XI. ENDPOINT GAUGE DATA AND BRANE HIGGSING

The toy operator in Sec. X becomes physical once its entries can be assigned to recognizable string or boundary data. Open strings and D-branes give one sharp language for this assignment. The route follows the original string-theory ambition: a particle spectrum is encoded in worldsheet and endpoint data, and spacetime fields arise as low-energy modes of that spectrum.

A. Endpoint labels as a source of matrix structure

An open string ending on a stack of N branes carries endpoint labels. In modern language these labels are Chan–Paton data. The corresponding low-energy fields sit in $N \times N$ matrices; for coincident branes, the massless excitations form a $U(N)$ gauge field and adjoint scalar fields on the brane worldvolume [2]. Polchinski’s review states the same logic in compactification language: current algebras of the compact worldsheet theory give space-time gauge bosons, and D-brane boundary conditions give open-string gauge fields along the brane [3].

For the present manuscript the important point is the matrix origin. Endpoint labels give a natural place for diagonal and off-diagonal channels:

$$|\psi; m, n\rangle, \quad m, n = 1, \dots, N, \quad (89)$$

with fields

$$A_a = (A_a)^m{}_n, \quad \Phi^I = (\Phi^I)^m{}_n. \quad (90)$$

The diagonal entries describe strings whose endpoints lie on the same brane. The off-diagonal entries describe strings stretched between distinct branes. At coincidence the off-diagonal vector modes complete the nonabelian gauge multiplet. This is precisely the type of two-channel bookkeeping needed by Eq. (11).

The source-backed part of the argument ends here. The DeVries-specific step is sharper and remains open: the endpoint algebra must produce the two entries

$$\kappa_J^2 = J, \quad \tau_J = J, \quad (91)$$

with $J = 3/4$ and $J = 2$ entering in the ordered electroweak way. The endpoint labels supply a matrix arena; selecting Eq. (91) requires additional dynamics.

B. Worldvolume Yang–Mills and adjoint scalars

The low-energy action for coincident D-branes contains the nonabelian Yang–Mills field strength and adjoint scalars [2]. Schematically,

$$S_{\text{brane}} = \int d^{p+1}\xi \, \text{Tr} \left[-\frac{1}{4g_{\text{YM}}^2} F_{ab} F^{ab} - \frac{1}{2} D_a \Phi^I D^a \Phi^I + \frac{1}{4} [\Phi^I, \Phi^J]^2 + \dots \right]. \quad (92)$$

The schematic form is enough for the present program. Gauge fields and scalars occupy the same matrix algebra. The covariant derivative

$$D_a \Phi^I = \partial_a \Phi^I + i[A_a, \Phi^I] \quad (93)$$

couples off-diagonal vector modes to scalar expectation values. When two branes are separated, the off-diagonal gauge fields acquire a mass proportional to the separation; equivalently, the mass is the energy of a string stretched between the branes [2]. The brane construction therefore gives a geometric Higgs mechanism with a scalar modulus and charged vector modes in one matrix system.

This structure motivates the endpoint route for the DeVries problem. The positive branch is meant to describe a vector pole-ratio clue. The negative branch asks for a scalar or order-parameter partner. A brane worldvolume action naturally contains both vector and scalar matrix entries. The target is a two-dimensional reduction normalized to

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = \tau_J = J. \quad (94)$$

C. Endpoint dictionary for the DeVries entries

The endpoint hypothesis can now be stated as a dictionary. Let h_J be the boundary order-parameter amplitude and a_J the endpoint-current amplitude. Let $\mathcal{H}_J = \text{span}\{h_J, a_J\}$ be the two-dimensional light sector. A candidate derivation must provide three pieces of data:

$$\text{zero reference: } \langle h_J | K_J | h_J \rangle = 0, \quad (95)$$

$$\text{trace entry: } \langle a_J | K_J | a_J \rangle = J, \quad (96)$$

$$\text{mixing entry: } \langle h_J | K_J | a_J \rangle \langle a_J | K_J | h_J \rangle = J. \quad (97)$$

Here K_J is the normalized quadratic kernel obtained after integrating out heavier world-volume or bulk modes. Eq. (96) should be traced to a current algebra, adjoint Casimir, or boundary kinetic term. Eq. (97) should be traced to an endpoint overlap, a brane-localized coupling, or a generator normalization. Eq. (95) fixes the baseline for the order-parameter channel.

The representation data suggest the following table:

DeVries entry	Endpoint or brane origin	Required derivation
$\tau_J = J$	adjoint/current trace, bound- ary kinetic term, or compact current algebra	show the normalized trace is the relevant quadratic invari- ant
$\kappa_J^2 = J$	endpoint overlap, stretched- string mixing, brane/order-parameter coupling	show the off-diagonal product or equals the same invariant
$\Sigma_{hh,J} = 0$	order-parameter channel	reference justify the subtraction by a symmetry, boundary condi- tion, or renormalization con- vention
$(J_H, J_{\text{adj}}) = (3/4, 2)$	doublet order parameter and adjoint current	derive the ordered W/Z sam- pling rule

This dictionary is restrictive. A completion must determine the entries, the basis, and the normalization from the source theory.

D. Scalar branch from brane separation

The brane picture also gives a natural language for the negative branch. In the D-brane Higgs mechanism, scalar expectation values describe brane positions. Off-diagonal vector fields become massive when the branes separate. Near coincidence, the off-diagonal modes become light and the full matrix structure controls the local physics [2]. Thus a two-branch matrix can plausibly pair a vector branch with a scalar or separation branch.

For the DeVries construction, this becomes the following target. There should be a scalar or boundary modulus φ_J and a normalized kernel K_J such that

$$K_J|_{\mathcal{H}_J} \sim \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad \varphi_J \text{ is read from } u_-(J). \quad (98)$$

Here $u_-(J)$ is the eigenvector of the negative eigenvalue. The phrase “read from” is an obligation: it must become a gauge-invariant map from the eigenvector data to a scalar quantity such as $H^\dagger H$, a brane separation, a boundary modulus, or a Higgs-potential parameter.

The scalar branch should also respect the electroweak ray. If the compact or brane model is meant to reproduce the Standard Model Higgs mechanism, the radial vacuum parameter v controls the common mass scale while the DeVries quotient controls the projective direction. A brane-separation interpretation must therefore identify which modulus supplies the radial scale and which normalized matrix supplies the angular data.

E. Endpoint completion criterion

The endpoint route makes the proposal more constrained. It asks for an explicit reduction:

$$\begin{aligned} \text{endpoint data} &\longrightarrow \text{worldvolume kernel} \longrightarrow \mathcal{M}_J, \\ \mathcal{M}_J &\longrightarrow \{x_+(J), x_-(J)\} \longrightarrow \text{pole ratio and scalar branch.} \end{aligned} \tag{99}$$

Each arrow has a check. The first arrow is source-supported by Chan–Paton labels, D-brane gauge fields, and adjoint scalars. The second arrow is the DeVries-specific determinant derivation. The third arrow is the exact algebra of Sec. II. The last arrow is the pole-placement and negative-branch problem.

This is also where the seventies lineage becomes substantive. The original dual amplitude began as a spectral object for hadron physics, and open-string amplitudes expose a tower of particle poles [2]. The present manuscript uses that lineage as method: start from a spectral clue, search for the worldsheet or boundary object that produces it, then translate the result into the electroweak pole scheme. The general string-theory idea that particle data can arise from endpoint and compactification spectra is source-backed; the electroweak translation is speculative.

XII. G_2 LOCALIZATION AND LOCAL SPECTRAL OPERATORS

The G_2 route supplies a third concrete language for the DeVries program. Its role in this manuscript is local. It gives a framework in which gauge fields, charged matter, order-parameter data, and deformation modes can live at singular loci of a compactification. M-theory on G_2 -holonomy spaces is a natural route to four-dimensional $\mathcal{N} = 1$ structure; singularities provide the gauge and matter sectors needed for particle physics [13–15]. The DeVries question is whether a localized G_2 operator can reduce to the same two-channel determinant used in the pole-ratio clue.

A. Singular support for gauge and matter sectors

The source-backed starting point is the localization of degrees of freedom. Witten explains that G_2 compactifications with singularities can support abelian or nonabelian gauge groups with chiral fermions, with anomaly inflow from the bulk supplying a consistency condition [13]. Acharya and Witten describe examples where singular loci carry nonabelian gauge groups and conical or intersection points support chiral multiplets [14]. Acharya and Gukov review the broader picture: singular G_2 spaces are the essential setting for nonabelian gauge groups, chiral fermions, and strongly coupled gauge dynamics in M-theory compactifications [15].

The DeVries problem asks for a local sector with three named pieces:

$$\left(\begin{array}{c} \text{singular gauge locus} \\ \text{localized charged matter or order parameter} \\ \text{deformation mode of the singularity} \end{array} \right) \rightsquigarrow \mathcal{M}_J. \quad (100)$$

The singular gauge locus is the natural address for adjoint/current data. The localized charged matter or order-parameter mode is the natural address for the doublet side. The deformation mode is the natural address for the scalar branch.

This local focus also controls the topology issue. A compact G_2 model imposes global constraints, including anomaly, representation, and line-operator data. The determinant can be tested locally first. A global compactification can then be asked to support the same local operator and the Standard Model global form.

B. Higgs-bundle variables on an associative three-cycle

Modern local G_2 gauge sectors can be organized by a Higgs-bundle-like system on an associative three-cycle M_3 . Braun, Cizel, Hubner, and Schafer-Nameki describe a gauge field together with a one-form Higgs field on M_3 , with matter localized at critical loci and interactions controlled by gradient-flow data [16]. In the abelianized local description, the Higgs field can be written schematically as

$$\Phi = \sum_{i=1}^n t_i df_i, \quad \rho = \sum_{i=1}^n t_i \rho_i, \quad \Delta f_i = \rho_i, \quad (101)$$

where t_i are Cartan generators and the ρ_i encode charge distributions. For a representation with charge vector $Q = (q_1, \dots, q_n)$, the effective Morse function is

$$f_Q = \sum_{i=1}^n q_i f_i, \quad \rho_Q = \sum_{i=1}^n q_i \rho_i. \quad (102)$$

Matter in the representation R_Q localizes at points where

$$df_Q(p) = 0. \quad (103)$$

Flow lines between such critical points can generate mass terms, and trivalent flow trees can generate Yukawa couplings when the representation charges sum to zero [16]. This is a strong local dictionary: gauge data, matter localization, mass terms, and interactions are all functions of the same geometric input.

For the DeVries construction, the Higgs-bundle variables give a possible address for the local operator:

$$\mathcal{O}_J^{G_2} = \mathcal{O}_J(W, \Phi, \rho_Q, f_Q), \quad (104)$$

where W is the gauge field on M_3 . The precise form of $\mathcal{O}_J^{G_2}$ is open. The variables W , Φ , and f_Q already separate adjoint gauge data, scalar deformation data, and localized representation data.

C. Local two-channel kernel

Let a_J be an adjoint/current fluctuation supported along the singular gauge locus, and let h_J be a localized order-parameter fluctuation at a critical point of f_Q . Let

$$\mathcal{H}_J^{G_2} = \text{span}\{h_J, a_J\}. \quad (105)$$

After integrating out heavier localized and bulk modes, the local inverse propagator or spectral kernel has the same abstract form as the endpoint and interval kernels:

$$K_J^{G_2}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{G_2}(\lambda) & \Sigma_{ha,J}^{G_2}(\lambda) \\ \Sigma_{ah,J}^{G_2}(\lambda) & \lambda + \Sigma_{aa,J}^{G_2}(\lambda) \end{pmatrix}. \quad (106)$$

The DeVries target is

$$\Sigma_{hh,J}^{G_2} = 0, \quad \Sigma_{aa,J}^{G_2} = J, \quad \Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J. \quad (107)$$

This target is a local statement. It can be tested near one singular locus or one conical point before a compact global model is specified.

D. Dictionary for the G_2 entries

The local G_2 dictionary is:

DeVries entry	G_2 origin	Derivation target
$\Sigma_{aa,J}^{G_2} = J$	ADE singular gauge sector, current algebra, or adjoint fluctuation along M_3	show that the normalized local trace is the relevant quadratic invariant
$\Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J$	localized matter overlap, flow-line coupling, bifundamental vev, or singularity-deformation mixing	derive the off-diagonal product from the local wavefunctions or flow data
$\Sigma_{hh,J}^{G_2} = 0$	reference order-parameter channel at a critical point of f_Q	justify the zero entry through local normalization or symmetry of the critical-point problem
$(J_H, J_{\text{adj}}) = (3/4, 2)$	localized doublet/order-parameter data and adjoint/current data	derive the ordered electroweak assignment inside the same local singularity model

This table turns G_2 geometry into a determinant problem. It also prevents the geometry from becoming a decorative completion. Each geometric object must provide one entry of $K_J^{G_2}$, one assignment rule, or one consistency check.

The first local test is therefore entry-by-entry. The adjoint entry should come from the gauge-field fluctuation W along the singular gauge locus, with the normalized ADE or adjoint invariant giving

$$W|_{M_3} \rightsquigarrow \Sigma_{aa,J}^{G_2} = J. \quad (108)$$

The off-diagonal product should come from the localized matter and Higgs-bundle data,

$$(\Phi, \rho_Q, f_Q) \rightsquigarrow \Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J, \quad (109)$$

through a normalized flow-line overlap, bifundamental vev, or singular deformation coupling.

The order-parameter reference should come from the critical-point normalization of f_Q ,

$$df_Q(p) = 0 \quad \rightsquigarrow \quad \Sigma_{hh,J}^{G_2} = 0. \quad (110)$$

This is the local G_2 analogue of the interval Schur-complement target: the source theory must provide the three named entries before the singularity language has derivational force.

E. Higgsing as singular deformation

Acharya and Witten give a pattern directly relevant to this manuscript: in examples where singular loci meet, gauge groups live on the singular components, chiral multiplets can be supported at the meeting point, and Higgsing corresponds to a geometric deformation or smoothing of the singularity [14]. In their brane-related examples, a vev for a bifundamental field can break a product gauge symmetry to a diagonal subgroup, while the local singular geometry changes accordingly.

For DeVries, this suggests the deformation chain

$$\begin{aligned} \text{intersecting singular loci} &\longrightarrow \text{localized chiral/order-parameter mode} \\ &\longrightarrow \text{deformed singularity} \longrightarrow K_J^{G_2}(\lambda). \end{aligned} \quad (111)$$

The scalar branch would then be read from the deformation direction. If $u_-(J)$ is the negative-branch eigenvector in $\mathcal{H}_J^{G_2}$, the target scalar map is

$$\begin{aligned} u_-(J) &\longmapsto \mathcal{F}_{G_2}, \\ \mathcal{F}_{G_2} &\in \left\{ \begin{array}{l} \text{deformation parameter, } f_Q(p), \rho_Q, \\ \text{bifundamental vev, localized modulus} \end{array} \right\}. \end{aligned} \quad (112)$$

The map must also respect the electroweak ray. The deformation parameter can supply the scalar side, while the pole-ratio branch supplies the projective vector side.

F. Anomaly and global checks

The G_2 route has strong consistency tests. Witten's anomaly-inflow discussion links local singularity data to four-dimensional chiral matter and anomaly cancellation [13]. Therefore a local DeVries operator has to pass the following checks:

1. the singular gauge locus supports the required adjoint sector;
2. localized charged matter has the desired representation content;
3. anomaly inflow or its local analogue is compatible with the charged fields;
4. the global gauge-group form of Sec. XV is compatible with the local representations;
5. flavor labels remain separate from compactification topology until a representation map is derived.

These checks can reject a candidate completion even when the two-channel determinant is obtained locally.

G. G_2 completion criterion

The G_2 route has the same completion chain as the endpoint and interval routes:

$$\begin{aligned} \text{singular } G_2 \text{ neighborhood} &\longrightarrow (W, \Phi, \rho_Q, f_Q) \longrightarrow K_J^{G_2}(\lambda), \\ K_J^{G_2}(\lambda) &\longrightarrow \lambda^2 + J\lambda - J \longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{G_2}, \mathcal{C}_{\text{top}}\}. \end{aligned} \tag{113}$$

The first arrow is source-backed by singular G_2 compactification and Higgs-bundle literature. The second arrow is the DeVries determinant derivation. The final arrow is the pole-placement, scalar-branch, and global-form assignment. A future proof can start locally, then extend to compact global data after the determinant appears.

XIII. COMPARISON OF THE THREE LOCAL ROUTES

The endpoint, interval, and G_2 sections introduce three ways to place the DeVries determinant inside a theory with localized gauge and order-parameter data. This section puts them in a common format. Each route is tested by the same normalized two-channel kernel, electroweak assignment, branch interpretation, and global compatibility conditions. A derivation in any route must supply a physical origin for the matrix entries and turn the string, extra-dimensional, or singular-geometric address into a calculation.

A. Common reduction problem

Let $\mathcal{S}_r$ denote one of the three route descriptions,

$$r \in \{\text{end}, \text{int}, G_2\}, \quad \mathcal{S}_r = \begin{cases} \mathcal{S}_{\text{end}} & \text{open-string endpoint or D-brane sector,} \\ \mathcal{S}_{\text{int}} & \text{interval Kaluza–Klein boundary sector,} \\ \mathcal{S}_{G_2} & \text{localized singularity sector.} \end{cases} \quad (114)$$

The desired reduction chooses a two-dimensional light subspace

$$\mathcal{H}_{J,r} = \text{span}\{h_{J,r}, a_{J,r}\}, \quad (115)$$

where $h_{J,r}$ is the order-parameter channel and $a_{J,r}$ is the adjoint/current channel. In this basis the reduced spectral kernel has the form

$$K_{J,r}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^r(\lambda) & \Sigma_{ha,J}^r(\lambda) \\ \Sigma_{ah,J}^r(\lambda) & \lambda + \Sigma_{aa,J}^r(\lambda) \end{pmatrix}. \quad (116)$$

The DeVries target is the simultaneous limit and normalization

$$\Sigma_{hh,J}^r = 0, \quad \Sigma_{aa,J}^r = J, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (117)$$

Equation (117) is stronger than a qualitative mechanism. It assigns one geometric, endpoint, or boundary object to each matrix entry. It also fixes the algebraic route to

$$\det K_{J,r}(\lambda) = \lambda^2 + J\lambda - J, \quad \lambda \in \{x_+(J), x_-(J)\}. \quad (118)$$

The reduction has three addresses. The endpoint address is supported by the open-string fact that endpoint labels and coincident branes generate matrix-valued gauge fields, with brane positions supplying scalar data [2, 3]. The interval address is supported by extra-dimensional models in which boundary conditions and brane-localized terms set vector spectra [7]. The G_2 address is supported by singular compactifications that localize gauge sectors and chiral matter, together with Higgs-bundle variables organizing local matter and interaction data [13–16]. These sources support the use of the addresses. They leave Eq. (117) as the shared derivation target.

B. Entry-by-entry comparison

The first entry is the diagonal current contribution,

$$a_{J,r} \rightsquigarrow \Sigma_{aa,J}^r = J. \quad (119)$$

The candidate source objects are a Chan–Paton or brane gauge amplitude, a boundary value of a bulk gauge mode, or a fluctuation along a singular gauge locus.

The second entry is the off-diagonal product,

$$h_{J,r} \leftrightarrow a_{J,r}, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (120)$$

The candidate source objects are a stretched open string or endpoint overlap, a boundary overlap between an order parameter and a bulk gauge profile, or a localized matter/deformation coupling.

The third entry is the order-parameter reference contribution,

$$h_{J,r} \rightsquigarrow \Sigma_{hh,J}^r = 0. \quad (121)$$

The candidate source objects are a boundary reference channel, a normalized boundary order-parameter self-energy, or a critical-point normalization of a localized mode. Appendix C 4 records the entry-by-entry proof obligations.

The following table records the three entries in a common dictionary.

Kernel entry	Endpoint/D-brane route	Interval KK route	G_2 route
$\Sigma_{aa} = J$	Chan–Paton current or bulk gauge amplitude	boundary ADE gauge fluctuation	or adjoint invariant
$\Sigma_{ha}\Sigma_{ah} = J$	stretched string or endpoint overlap	order-parameter/bulk-profile overlap	localized matter or deformation coupling
$\Sigma_{hh} = 0$	order-parameter reference channel	normalized scalar term	boundary critical-point normalization

C. Electroweak assignment as a route-independent target

The same electroweak assignment target applies to every route. The low-energy field-theory data contain an $SU(2)_L$ Higgs doublet and adjoint gauge currents. Their natural

quadratic invariants give

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (122)$$

The target map is

$$\mathcal{A}_r : (\mathcal{S}_r, \mathcal{H}_{J,r}) \longrightarrow (J_W, J_Z) = (J_H, J_{\text{adj}}). \quad (123)$$

This map is a theorem target because the observed W and Z vector bosons are both gauge-sector particles after symmetry breaking. The route must explain why the charged comparison samples the order-parameter invariant while the neutral comparison samples the adjoint/current invariant. The required derivation must produce the ordered pair from the gauge-Higgs coupling structure, from endpoint boundary labels attached to the Higgsing field, from interval boundary data that distinguish charged and neutral sectors, or from a localized singularity model where doublet and adjoint data meet.

If the ordered assignment and pole-placement theorem hold, the pole-ratio reading has the common form

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} \quad (124)$$

after the pole scheme of Sec. III has been fixed and the O8 matching theorem in Sec. XIIE has been proved. This equation states the desired low-energy placement. A route that naturally yields the same quotient at a compactification scale or unification scale would locate the algebra in a source theory. The low-energy pole statement would then require the matching theorem as an additional result.

D. Branch interpretation and scalar data

Every route diagonalizes the same two-channel kernel. Therefore each route has to account for both eigenvectors. Let

$$u_{\pm,J,r} = \alpha_{\pm,J,r} h_{J,r} + \beta_{\pm,J,r} a_{J,r} \quad (125)$$

be normalized eigenvectors of the route kernel in the DeVries limit. The positive branch supplies the pole-ratio clue. The negative branch supplies the scalar-side question,

$$u_{-,J,r} \rightsquigarrow \mathcal{F}_{\text{sc},r}. \quad (126)$$

The functional $\mathcal{F}_{\text{sc},r}$ depends on the route. Endpoint, interval, and G_2 sources provide candidate scalar addresses: brane or boundary moduli, localized scalar modes, and singularity-deformation data. These addresses acquire physical content after a gauge-invariant functional, normalization, and electroweak scheme are specified.

The scalar map has to be tied to the electroweak ray. The gauge-Higgs system has a radial vacuum scale and a projective vector-mass ratio. The two-root structure is physically restrictive because a determinant used for the pole clue also supplies the partner branch. A source route must identify how the positive branch fixes the projective vector datum and how the negative branch enters the radial or scalar sector.

E. Scale placement

The route comparison separates three scale placements:

$$\text{pole scale,} \quad \text{compactification or boundary scale,} \quad \text{unification scale.} \quad (127)$$

This paper takes the pole scale as the proposed attachment point for the electroweak clue. This choice gives the clue its direct electroweak content. The GUT lineage supplies the comparison class: a clean weak-angle value can be a high-scale structural datum, then acquire low-energy meaning through running and thresholds. In the present construction the clean value is attached to the pole ratio, so the scale-placement theorem is logically prior to any compactification geometry.

The common matching theorem has the form

$$\mathcal{S}_r \longrightarrow K_{J,r}^{\text{bare}}(\lambda) \longrightarrow \Gamma_{\text{eff}}^{(4)}[H, W, B] \longrightarrow \Delta_{V,T}^{-1}(s) \Big|_{s=s_{W,Z}}. \quad (128)$$

This chain is the open O8 theorem. The first arrow is the route reduction. The second arrow matches the source kernel into a four-dimensional gauge-Higgs effective action. The third arrow places the result in the dressed transverse pole convention of Sec. III. Endpoint, interval, and G_2 routes differ in the first arrow; they share the last two arrows when the pole reading is retained. Appendix C 2 gives the equivalent transverse self-energy form of the same theorem.

For endpoint and interval routes, the most direct challenge is the relation between a boundary determinant and a four-dimensional pole. Boundary conditions usually define a

spectrum before loop corrections and before the conversion to pole observables. The route must explain how the determinant survives matching to the pole definition, or how the pole reading emerges from a dressed boundary self-energy.

For the G_2 route, the challenge is sharper. A local singularity model naturally lives at a compactification or high scale. The local kernel can organize representations and couplings, yet a pole-spectrum reading requires a low-energy matching chain. The resulting target is a two-stage theorem:

$$\mathcal{S}_{G_2} \longrightarrow K_{J,G_2} \longrightarrow \Delta_{V,T}^{-1}(s) \quad \text{with} \quad s = s_{W,Z}. \quad (129)$$

The first arrow is local geometry. The second arrow is four-dimensional matching. Both arrows remain open obligations.

F. Route outcome

A derivation in any route must satisfy the common kernel target, the ordered electroweak assignment, the pole-matching chain, the scalar-branch map, and the global compatibility checks. Appendix C9 collects the full checklist. Extra light channels, a different trace, a different mixing product, a high-scale placement, or an incompatible global form would classify the DeVries relation inside a larger family of spectral constructions.

XIV. FLAVOR BOUNDARY AND REPRESENTATION ALGEBRA

The electroweak mass-ratio problem and the flavor problem carry separate obligations in this manuscript. The working boundary condition is:

$$\begin{aligned} &\text{DeVries electroweak determinant} \\ &\quad \text{and generation/flavor organization} \\ &\text{are linked only by an explicit model.} \end{aligned} \quad (130)$$

This condition requires any compactification or endpoint picture to give an explicit equation linking flavor data to the electroweak determinant.

Representation theory remains essential. In the Standard Model and in the familiar grand-unification lineage, fermions live in representations of the internal gauge group, while

gauge bosons live in the adjoint sector [19, 20]. $SU(5)$, $Spin(10)$, Pati–Salam, and E_6 show how a larger algebra can unify the Standard Model representation data. This is the correct role of the GUT material in the present paper: it supplies representation discipline and a comparison class for structural weak-angle statements.

The DeVries construction uses a different representation question. The electroweak determinant asks why one sample is associated with the Higgs/order-parameter doublet invariant and the other with the adjoint/current invariant:

$$J_H = T_H(T_H + 1) = 3/4, \quad J_{\text{adj}} = T_{\text{adj}}(T_{\text{adj}} + 1) = 2. \quad (131)$$

Flavor representations can constrain a completion. The missing step is a derived map from endpoint, compactification, or localized matter data into the two-channel electroweak operator.

The $SO(32)$ -flavor idea should therefore be treated as an organizing boundary sector. In an endpoint language it may label external or internal flavor degrees of freedom, Chan–Paton-like data, or a bookkeeping algebra for fermion multiplicities. Its required checks are concrete:

1. identify the representation space carrying the $SO(32)$ -flavor labels;
2. map the Standard Model chiral fermions into that space;
3. show anomaly cancellation and hypercharge normalization;
4. state the relation to the global quotient $\Gamma \subset \mathbb{Z}_6$;
5. couple the flavor sector to the electroweak determinant only through a derived operator.

This keeps the flavor idea as a compatibility sector. The electroweak ratio concerns W/Z pole structure and the Higgs/order-parameter channel. Generation structure concerns chiral matter, Yukawa textures, and possible endpoint or compactification multiplicities. A string/Kaluza–Klein completion may eventually bind them together through an equation or a local geometric construction.

The compactification version has an additional caveat. Geometry can generate gauge symmetries, chiral localization, and multiple wavefunctions with the same gauge quantum

numbers [12, 14, 16]. That makes topology relevant to flavor. The manuscript still treats generation counting as a separate task because the DeVries determinant is a two-channel electroweak object. The decisive test is whether the same local model that yields generations also yields

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J \quad (132)$$

with the ordered electroweak samples in Eq. (131).

The practical rule for later sections is therefore simple. Flavor material may enter as a constraint on endpoints, compactification, anomaly cancellation, and global form. It may motivate a local operator. The W/Z mass-ratio argument rests on the electroweak determinant and its pole placement.

XV. GLOBAL FORM OF THE STANDARD MODEL GAUGE GROUP

The local Standard Model gauge algebra admits several global forms. The gauge group may be written as

$$G_{\text{SM}} = \frac{SU(3)_C \times SU(2)_W \times U(1)_Y}{\Gamma}, \quad \Gamma \subset \mathbb{Z}_6, \quad (133)$$

with $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$ as the standard possibilities [17, 18]. Local perturbative fields see the Lie algebra. Line operators, magnetic sectors, theta-angle periodicities, and topological probes see the quotient.

This is relevant to the DeVries program for two reasons. First, any compactification, endpoint, or G_2 completion has to specify the global group it realizes. Second, a pole-spectrum determinant for the electroweak sector has to coexist with the same line-operator spectrum as the low-energy Standard Model. Thus a proposed $\mathcal{O}_J$ comes with a global-form compatibility condition:

$$\mathcal{O}_J \text{ is defined in a theory with } G_{\text{SM}} = (SU(3) \times SU(2) \times U(1))/\Gamma. \quad (134)$$

Tong's line-operator analysis classifies how the quotient changes the electric, magnetic, and dyonic line sectors [17]. The quotient restricts admissible electric line operators and changes the allowed magnetic line operators through Dirac quantization. It also changes theta-angle periodicities. After electroweak symmetry breaking, the line sectors descend to electromagnetic probes, so Γ leaves a low-energy topological signature.

Hsin and collaborators propose a transport-style probe of this global form through a fractional topological response coefficient [18]. The important point for this manuscript is structural: the global form is, in principle, observable through nonlocal or topological response data. It supplies a completion constraint for the DeVries program, because a string or compactification model generally determines more than a local Lie algebra.

Grand unification provides the comparison class. $SU(5)$ and $Spin(10)$ naturally package Standard Model representations in a way that favors the $\mathbb{Z}_6$ quotient, and the subgroup $S(U(2) \times U(3))$ realizes the quotient structure inside $SU(5)$ [19, 20]. This places the GUT lineage beside the DeVries pole reading: high-scale structural placement is already known in unified models, while the DeVries construction assigns its clean value to the low-energy pole ratio and therefore needs a separate pole-placement derivation.

The global-form constraint can be expressed as a commutative diagram requirement. If a compactification or endpoint model gives an ultraviolet group G_{UV} , then the low-energy electroweak sector should fit into

$$\begin{array}{ccc} G_{UV} & \longrightarrow & G_{SM} \\ \downarrow & & \downarrow \\ \mathcal{O}_{UV} & \longrightarrow & \mathcal{O}_J, \end{array} \tag{135}$$

where the upper arrow fixes Γ and the lower arrow reduces the ultraviolet spectral operator to the DeVries two-channel operator. A model that supplies $\mathcal{O}_J$ while leaving Γ unspecified has an incomplete topological sector.

The mass-ratio argument itself should use local electroweak fields and pole positions. The global-form material acts as a filter on completions. A viable string/Kaluza–Klein/ G_2 derivation should state:

1. which quotient Γ appears;
2. which line operators survive compactification and electroweak breaking;
3. how the Higgs/order-parameter field transforms under the chosen global group;
4. how the DeVries operator respects the same quotient.

XVI. CONCLUSION AND ANALYTICAL STATUS

The construction begins with an exact two-branch algebra. The first assumption $A1$ fixes

$$Q(J) = \mu^2 \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad J = s(s+1),$$

whose dimensionless eigenvalues obey $x^2 + Jx - J = 0$. The quotient

$$1 - \frac{x_+(3/4)}{x_+(2)}$$

is therefore a rigid output of $A1$ after the ordered samples $(3/4, 2)$ are chosen. The mathematical statement is derived. The physical statement begins with the assignment and placement hypotheses.

The placement hypothesis $A2$ uses the complex-pole mass ratio

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}.$$

This choice gives the spectral language a low-energy meaning and separates the proposal from $\overline{\text{MS}}$ running angles, effective weak angles, Breit–Wigner line-shape masses, and high-scale GUT boundary values. The pole placement is an assumption awaiting a dynamical pole condition. Appendix C 2 states the corresponding theorem target.

The Standard Model gauge-Higgs sector supplies the projective datum

$$\frac{M_W^2}{M_Z^2} = \frac{g^2}{g^2 + g'^2}$$

and the broken-to-unbroken vacuum ray. It also supplies two representation data relevant to the proposed assignment: the Higgs/order-parameter doublet with $J_H = 3/4$, and the adjoint/current channel with $J_{\text{adj}} = 2$. The assignment hypothesis $A3$ is the ordered map

$$(J_W, J_Z) = (J_H, J_{\text{adj}}) = (3/4, 2)$$

from gauge-Higgs, boundary, endpoint, or compactification data. The first theorem target $T1$ is a derivation of this map from the same source data that define the two-channel operator.

The negative branch belongs to the same diagonalization as the positive branch. Its scalar interpretation remains conditional on a gauge-invariant functional, a normalization, and a scheme. Candidate targets include a Higgs-potential parameter, a scalar pole, a boundary modulus, a brane separation mode, or a compactification deformation. This branch is

retained because a source theory that derives the positive-branch determinant must also account for the partner eigenvalue.

The second theorem target $T2$ is the route-kernel derivation. Endpoint, interval, and G_2 sources give addresses for a two-channel kernel. In common notation the required reduction is

$$K_J(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J} & \Sigma_{ha,J} \\ \Sigma_{ah,J} & \lambda + \Sigma_{aa,J} \end{pmatrix}, \quad \Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J}\Sigma_{ah,J} = J.$$

Open-string endpoint data, interval boundary conditions, and localized G_2 gauge sectors provide source-backed addresses for such a kernel. The exact DeVries entries remain to be derived from one source theory with a specified field basis and inner product.

The third theorem target $T3$ is the scalar interpretation of the partner branch. The present paper establishes the algebraic clue, the pole-placement hypothesis, and source-backed addresses for the determinant. A dynamical claim for the DeVries quotient requires the two-channel determinant, the ordered electroweak assignment, the pole-matching theorem, and the controlled scalar interpretation of the negative branch. Global-form and representation checks then constrain any compact completion. Appendix C collects these requirements in theorem-target form.

Appendix A: Algebraic details

For completeness, we record the elementary algebra. From Eq. (5),

$$x_{\pm}(J) = \frac{-J \pm \sqrt{J(J+4)}}{2}. \quad (\text{A1})$$

The positive branch has the large- J expansion

$$x_+(J) = 1 - \frac{1}{J} + \frac{2}{J^2} + O(J^{-3}), \quad (\text{A2})$$

while

$$x_-(J) = -J - 1 + \frac{1}{J} + O(J^{-2}). \quad (\text{A3})$$

Thus the positive branch approaches a fixed asymptotic value while the negative branch carries the large negative partner required by the trace.

For $J = 3/4$,

$$x_+\left(\frac{3}{4}\right) = \frac{-3/4 + \sqrt{57}/4}{2} = \frac{\sqrt{57} - 3}{8}. \quad (\text{A4})$$

For $J = 2$,

$$x_+(2) = \frac{-2 + \sqrt{12}}{2} = \sqrt{3} - 1. \quad (\text{A5})$$

Therefore

$$s_{\text{dV}}^2 = 1 - \frac{\sqrt{57} - 3}{8(\sqrt{3} - 1)}. \quad (\text{A6})$$

Appendix B: Numerical reproducibility

The script ‘calculations/devries_spectrum.py’ computes the branch roots and the derived mass value from a supplied M_Z . The central functions are

```
x_plus(J) = (sqrt(J*J + 4*J) - J)/2
sin2_devries = 1 - x_plus(3/4)/x_plus(2)
MW = MZ * sqrt(1 - sin2_devries)
```

The test file ‘calculations/tests/test_devries_spectrum.py’ checks the root identities, the numerical value of s_{dV}^2 , and the seed value obtained from $M_Z = 91.1880 \text{ GeV}$.

Appendix C: Theorem targets for a derivation

This appendix records the analytical obligations in theorem form. The purpose is operational. Each target names its hypotheses, conclusion, current evidence, and failure mode. A complete version of the manuscript should either prove these statements, replace them with stronger statements, or demote the dependent claims.

1. Common format

The targets share a common data structure. Let $\mathcal{S}$ be a source theory or effective description. It may be a four-dimensional electroweak theory, a boundary effective action from an interval compactification, an open-string endpoint sector, or a localized G_2 system. Let $\mathcal{E}_J$ be the two-dimensional light subspace that is meant to carry the order-parameter and

current amplitudes. The desired reduction has the form

$$\mathcal{S} \longrightarrow \mathcal{E}_J \longrightarrow \mathcal{M}_J = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} \longrightarrow \{x_+(J), x_-(J)\}. \quad (\text{C1})$$

The reduction is meaningful after the normalization of $\mathcal{E}_J$ and the scheme in which the resulting eigenvalues are read have been specified. The pole-observable section fixes the electroweak comparison as a complex-pole statement, with Breit-Wigner/on-shell and running quantities treated as translated inputs [10, 11]. The electroweak section fixes the low-energy field-theory data that any assignment theorem must respect [5, 8, 9]. The string and Kaluza–Klein sections list candidate sources of the two-channel determinant [2, 3, 7].

For each target below, “proof” means an explicit derivation from a stated action, boundary condition, operator algebra, or compactification mode equation. A plausibility argument may motivate the target, yet the manuscript’s final physical claim must track the proven statement.

2. Target I: pole placement

Hypotheses. Assume a source theory $\mathcal{S}$ gives a two-branch spectral equation with roots $x_{\pm}(J)$. Assume the electroweak comparison is made with transverse gauge-boson complex poles,

$$\det \Delta_{V,T}^{-1}(s_V) = 0, \quad s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \quad (\text{C2})$$

Assume the mass and width symbols are in the pole parameterization of Eq. (C2). If an input uses the alternative parameterization $s_{\text{pole}} = (M - i\Gamma/2)^2$, first translate it to Eq. (C2). If W/Z inputs are quoted in a common variable-width Breit–Wigner line-shape convention, convert

$$M_{V,\text{pole}} = \frac{M_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}, \quad \Gamma_{V,\text{pole}} = \frac{\Gamma_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}, \quad \gamma_V = \frac{\Gamma_{V,\text{BW}}}{M_{V,\text{BW}}}. \quad (\text{C3})$$

This gives

$$M_{V,\text{BW}}^2 = M_{V,\text{pole}}^2 + \Gamma_{V,\text{pole}}^2. \quad (\text{C4})$$

Assume common experimental mass conventions are converted into the pole convention before forming

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (\text{C5})$$

Conclusion. The DeVries branch quotient is a pole-spectrum datum:

$$s_{\text{pole}}^2 = 1 - \frac{x_+(J_H)}{x_+(J_{\text{adj}})} \quad (\text{C6})$$

up to stated radiative, threshold, and scheme translations. The theorem must also explain why the assigned scale is the low-energy pole spectrum, with high-scale boundary placements handled by a separate derivation. For any compactification, endpoint, interval, or local G_2 source, the scale-placement statement has the matching form

$$\mathcal{S} \longrightarrow K_J^{\text{bare}}(\lambda) \longrightarrow \Gamma_{\text{eff}}[H, W, B] \longrightarrow \Delta_{V,T}^{-1}(s) \Big|_{s=s_{W,Z}}. \quad (\text{C7})$$

The first arrow is the source reduction. The second arrow is four-dimensional effective matching. The third arrow is the dressed transverse pole condition. The pole reading requires all three arrows in one scheme. Equivalently, the source kernel must produce the relevant pole self-energy term,

$$K_J^{\text{bare}}(\lambda) \longrightarrow \Pi_{T,V}^{(4)}(s; J), \quad \Delta_{V,T}^{-1}(s_V; J) = 0, \quad (\text{C8})$$

with the same pole parameterization and field basis used in Eq. (C2).

Current evidence. The manuscript already separates complex poles, Breit-Wigner/on-shell masses, $\overline{\text{MS}}$ running angles, effective weak angles, GUT boundary values, and compactification eigenvalues. Martin–Robertson supply the scheme language used in Sec. III. The CDF-II result has descriptive displacement status here.

Failure mode. The target fails if the derived clean value lives at a high-scale boundary, a running $\overline{\text{MS}}$ angle, an effective leptonic angle, or a compactification eigenvalue with a different translation law. In that case the pole-ratio reading becomes a motivating clue, and the manuscript relocates the physical statement to the derived scheme and scale. A compactification-scale or unification-scale derivation would still be valuable; it would convert the DeVries quotient into a boundary datum whose relation to W/Z pole data is a separate running, threshold, and matching problem.

3. Target II: electroweak assignment

Hypotheses. Let H be the Standard Model Higgs doublet and let $J_H = 3/4$ be its $SU(2)_L$ quadratic invariant. Let $J_{\text{adj}} = 2$ be the corresponding adjoint invariant for $SU(2)_L$ currents.

The gauge-Higgs sector contains the usual mass terms

$$M_W^2 = \frac{g^2 v^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2 = 0, \quad (\text{C9})$$

with the custodial structure and $\rho = 1$ tree-level relation discussed in Sec. IV.

Conclusion. There exists a natural map from gauge-Higgs data to ordered samples,

$$\mathcal{D}_{\text{EW}} : (H, J_\mu^a, B_\mu) \mapsto (J_W, J_Z) = (J_H, J_{\text{adj}}) = (3/4, 2), \quad (\text{C10})$$

such that

$$\frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}. \quad (\text{C11})$$

The derivation must say why the charged comparison samples the order-parameter invariant while the neutral comparison samples the adjoint/current invariant.

Current evidence. The Higgs doublet and adjoint invariants are fixed representation-theoretic facts. The electroweak mass matrix, photon null direction, custodial relation, and projective reading of $g : g'$ are standard low-energy data [5, 8, 9]. The manuscript has converted the assignment into a theorem target.

Failure mode. The target fails if the assignment uses observed vector spin or an arbitrary post-selection of J values. A successful derivation must arise from the gauge-Higgs variables, a boundary operator coupled to them, or a compactification mechanism whose low-energy reduction yields the ordered pair.

4. Target III: two-channel boundary determinant

Hypotheses. Let h_J be an order-parameter amplitude and a_J an adjoint/current amplitude in a light boundary subspace. Let their normalized quadratic kernel be

$$K_J(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J} & \Sigma_{ha,J} \\ \Sigma_{ah,J} & \lambda + \Sigma_{aa,J} \end{pmatrix}. \quad (\text{C12})$$

The source may be an interval theory with brane-localized terms, an open-string endpoint sector, or a localized singularity construction. Boundary mass and kinetic terms can alter eigenvalue conditions in extra-dimensional models [7]. Open-string endpoints and D-brane gauge sectors supply a separate endpoint language [2, 3].

Conclusion. There is a controlled limit and normalization in which

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J}\Sigma_{ah,J} = J, \quad (\text{C13})$$

so that

$$\det K_J(\lambda) = \lambda^2 + J\lambda - J. \quad (\text{C14})$$

Equivalently, the physical derivation fixes the two named entries

$$\tau_J = J, \quad \kappa_J^2 = J. \quad (\text{C15})$$

Current evidence. Sec. X gives the target action and the required entries. The literature sources support the existence of boundary-condition and endpoint mechanisms, while the exact DeVries entries remain conjectural.

Failure mode. The target fails if boundary variation gives a different trace, a different off-diagonal product, or an additional channel that changes the determinant class. The result in that case would be a larger family of secular relations containing the DeVries matrix as a limiting or approximate case.

5. Target IV: negative branch as scalar data

Hypotheses. Assume Targets I–III identify a physical two-channel matrix whose eigenvalues are $x_{\pm}(J)$. Let $u_{\pm}(J)$ be normalized eigenvectors. The negative branch is then part of the same diagonalization as the positive branch:

$$\mathcal{M}_J u_{\pm}(J) = x_{\pm}(J) u_{\pm}(J). \quad (\text{C16})$$

The scalar side of the electroweak theory contains the gauge-invariant radial datum, the Higgs pole, and the parameters of the symmetry-breaking potential [8, 9].

Conclusion. There exists a gauge-invariant scalar functional $\mathcal{F}_{\text{sc}}$ such that

$$\mathcal{F}_{\text{sc}}(H^{\dagger}H, D_{\mu}H, \text{ boundary data}) = C |x_{-}(J_{\star})| \quad (\text{C17})$$

for a stated $J_{\star}$, normalization C , and scheme. Candidate readings include a Higgs-potential parameter, the radial vacuum scale, a boundary modulus, or a brane-separation mode.

Current evidence. Sec. V records the scalar target table and the eigenvector interpretation. The negative branch is treated as an analytical partner of the pole clue, with its Higgs/order-parameter meaning open.

Failure mode. The target fails if the negative branch has purely auxiliary meaning in the derived source theory. The positive-branch clue can still motivate the determinant problem, while the scalar claims must then be removed from the physical conclusion.

6. Target V: global-form and localization compatibility

Hypotheses. Assume a string, brane, Kaluza–Klein, or G_2 source derives Targets II–IV. The low-energy gauge group has global form

$$G_{\text{SM}} = \frac{SU(3) \times SU(2) \times U(1)}{\Gamma}, \quad \Gamma \subseteq \mathbb{Z}_6. \quad (\text{C18})$$

Line operators and global identifications provide tests of the Standard Model gauge group beyond the Lie algebra [17, 18]. Singular G_2 spaces can localize gauge and chiral matter sectors in M-theory constructions [13–16].

Conclusion. The derived operator $\mathcal{M}_J$ is compatible with the allowed line operators, representation assignments, anomaly constraints, and localized matter data of the chosen completion. A successful completion must specify whether flavor labels are endpoint data, compactification data, representation-sector data, or a combination constrained by the source geometry.

Current evidence. Secs. XIV and XV state the current boundary conditions: the $\text{SO}(32)$ -flavor idea is a representation-sector clue, and global-form data are completion constraints. The manuscript avoids identifying generations with compactification topology until a representation map is derived.

Failure mode. The target fails if the determinant arises in a gauge or flavor sector incompatible with Standard Model line operators, anomaly conditions, or observed representations. A failure here would be decisive for a proposed completion even if the local two-channel determinant is mathematically correct.

7. Target VI: dimensional interpolation

Hypotheses. Assume the electroweak order parameter supplies an interpolation parameter t , represented in a source theory by compact, boundary, brane, or singular data. The full-gauge endpoint dictionary is

$$(D, n_{\text{KK}}) = (11, 7) : SU(3) \times SU(2) \times U(1), \quad (D, n_{\text{KK}}) = (9, 5) : SU(3) \times U(1)_{\text{em}}. \quad (\text{C19})$$

The colourless electroweak dictionary is

$$(D, n_{\text{KK}}^{\text{ew}}) = (7, 3) : SU(2) \times U(1), \quad (D, n_{\text{KK}}^{\text{ew}}) = (5, 1) : U(1)_{\text{em}}. \quad (\text{C20})$$

At the middle point $t = t_*$, assume a light subspace

$$\mathcal{H}_J(t_*) = \text{span}\{h_J, a_J\} \quad (\text{C21})$$

with h_J an order-parameter channel and a_J an adjoint/current channel.

Conclusion. The middle-dimensional sector defines a source operator

$$\mathcal{O}_J(t_*)|_{\mathcal{H}_J} \longrightarrow K_J(t_*, \lambda) = \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix}, \quad (\text{C22})$$

and the same sector yields the ordered electroweak samples

$$(J_W, J_Z) = (J_H, J_{\text{adj}}) = \left(\frac{3}{4}, 2\right). \quad (\text{C23})$$

The full-gauge middle line is then $(D, n_{\text{KK}}) = (10, 6)$. The colourless middle line is $(D, n_{\text{KK}}^{\text{ew}}) = (6, 2)$. The theorem also supplies the pole-matching chain in Eq. (C7).

Current evidence. Witten’s Kaluza–Klein analysis supplies the source trail for the seven-extra-dimensional $SU(3) \times SU(2) \times U(1)$ endpoint and the associated fermion quantum-number obstruction [12]. The same paper contains a five-sphere $SU(3) \times U(1)$ symmetry address. The six-dimensional string sources support a landscape of six-dimensional string settings and dualities [21–24]. Appendix D converts these facts into a source-status table and a middle-line reconstruction target.

Failure mode. The target fails if the middle-dimensional sector produces extra light channels, a kernel outside the DeVries determinant class, a high-scale datum with a different matching law, or an ordered assignment other than $(J_H, J_{\text{adj}}) = (3/4, 2)$. Such an outcome would classify the dimensional idea as an organizing analogy for a different source-theory reduction.

8. Claim hierarchy and acceptance criteria

The theorem targets separate four kinds of statements:

Level	Manuscript content	Required evidence
Derived	algebra of $Q(J)$, roots, branch identities, trace and determinant form	equation-level derivation in Sec. II
Source-supported	electroweak mass matrix, cited local source fragments and pole/running scheme distinctions, endpoint/brane, KK, and G_2 background	distinct references
Conjectural	pole placement of the DeVries quotient, negative-branch scalar reading, route-specific kernel entries	theorem target with assumptions and failure mode
Open	ordered J -assignment, full determinant derivation, scalar functional, boundary condition, endpoint algebra-form compatibility of a compact completion	proof from a stated action, boundary condition, endpoint algebra, or localized operator

A minimal derivation has the chain

$$\mathcal{S} \longrightarrow \mathcal{H}_J \longrightarrow K_J(\lambda) \longrightarrow (J_H, J_{\text{adj}}) \longrightarrow (s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}). \quad (\text{C24})$$

The source theory $\mathcal{S}$ may be a boundary effective action, an endpoint sector, or a localized G_2 system. The light subspace $\mathcal{H}_J$ must contain an order-parameter channel and an adjoint/current channel. The kernel $K_J(\lambda)$ must reduce to the DeVries determinant in a stated normalization. The ordered pair $(J_H, J_{\text{adj}}) = (3/4, 2)$ must arise from gauge-Higgs or source-theory data. The final arrow must attach the positive branch to the pole observable and the negative branch to a defined scalar functional.

The common acceptance criterion is

$$\det K_J(\lambda) = \lambda^2 + J\lambda - J, \quad (J_W, J_Z) = (3/4, 2), \quad u_-(J) \mapsto \mathcal{F}_{\text{sc}}. \quad (\text{C25})$$

Partial results still sharpen the construction when they prove one arrow in Eq. (C24). An interval model may derive the determinant class, an endpoint model may derive the ordered representation pair, and a G_2 model may supply a localization mechanism. Each result changes the status of a corresponding theorem target.

9. Derivation checklist

The theorem targets lead to the following checklist for future revisions:

1. State the source theory $\mathcal{S}$ and the light subspace $\mathcal{E}_J$.
2. Specify the normalization that makes $\Sigma_{hh,J} = 0$, $\Sigma_{aa,J} = J$, and $\Sigma_{ha,J}\Sigma_{ah,J} = J$.
3. Derive the ordered assignment $(J_H, J_{\text{adj}}) = (3/4, 2)$ from low-energy gauge-Higgs or boundary data.
4. Give the pole-scheme translation for every quoted W/Z mass input.
5. Identify the negative branch as scalar data, auxiliary data, or a derived sector with another physical role.
6. Check global-form, anomaly, and representation compatibility.

This checklist is the manuscript’s current definition of a complete derivation.

Appendix D: Dimensional interpolation target

This appendix records a dimensional version of the Higgs-sector problem. It is a theorem target, added here because it gives a compact way to organize the Kaluza–Klein, string, Higgs, and top-quark questions.

1. Full-gauge count

The full-gauge count keeps colour inside the compactification symmetry. Witten’s realistic Kaluza–Klein analysis gives the source-backed endpoint: a compact space with $SU(3) \times SU(2) \times U(1)$ symmetry requires seven extra dimensions in the class considered there, giving total dimension $D = 11$ [12]. The same source records the associated fermion quantum-number obstruction for that seven-dimensional compactification.

The proposed interpolation is

$$\begin{aligned}
D = 11, \quad n_{\text{KK}} = 7 & : SU(3) \times SU(2) \times U(1) \quad \text{massless-Higgs or unbroken endpoint,} \\
D = 10, \quad n_{\text{KK}} = 6 & : \text{DeVries interpolation sector,} \\
D = 9, \quad n_{\text{KK}} = 5 & : SU(3) \times U(1)_{\text{em}} \quad \text{infinitely broken endpoint.}
\end{aligned} \tag{D1}$$

The source status of the three lines is uneven, and the distinction is part of the claim. The source-backed data and reconstruction targets are:

Line	Source-backed content	DeVries obligation
$D = 11, n_{\text{KK}} = 7$	Witten's $SU(3) \times SU(2) \times U(1)$ define the massless-Higgs endpoint KK endpoint and its fermion and its relation to the electroweak quantum-number obstruction	ray
$D = 10, n_{\text{KK}} = 6$	reconstruction target	identify the six-dimensional inter- nal object, its light two-channel sector, and its kernel
$D = 9, n_{\text{KK}} = 5$	Witten's five-sphere discussion derive the infinite-Higgs or in- contains an $SU(3) \times U(1)$ sym- finitely broken endpoint with metry address	$U(1)_{\text{em}}$ as the surviving elec- troweak factor

The middle line is therefore the active theorem target. A derivation must identify the six-dimensional internal object, its boundary or compact operator, and its relation to the two-channel DeVries determinant.

2. Colourless electroweak count

There is a reduced count in which colour is treated as a spectator sector. The electroweak part of the endpoint symmetry then changes as

$$\begin{aligned}
D = 7, \quad n_{\text{KK}}^{\text{ew}} = 3 & : SU(2) \times U(1), \\
D = 6, \quad n_{\text{KK}}^{\text{ew}} = 2 & : \text{DeVries electroweak interpolation sector,} \\
D = 5, \quad n_{\text{KK}}^{\text{ew}} = 1 & : U(1)_{\text{em}}.
\end{aligned} \tag{D2}$$

This version makes the six-dimensional middle line especially natural as a string-theory address. Six-dimensional superstring vacua and dualities are a standard part of the source landscape: type IIA compactified on K3 and the heterotic string compactified on T^4 provide a six-dimensional string-string duality setting [21, 22]; heterotic small instantons give nonperturbative six-dimensional dynamics [23]; and six-dimensional anomaly and string-universality constraints provide a modern consistency framework [24]. These sources support the existence and rigidity of six-dimensional string settings. The DeVries kernel, ordered J -assignment, and pole placement remain separate reduction targets.

3. Interpolation parameter

The interpolation parameter must be stated in field-theory or geometric terms. The electroweak field theory supplies a radial order parameter through the Higgs vacuum expectation value. The compactification descriptions supply radii, boundary conditions, brane-localized terms, or singularity deformations. A viable derivation must identify one variable t and a family of source operators

$$\mathcal{O}_J(t), \quad t = 0 : SU(3) \times SU(2) \times U(1), \quad t = \infty : SU(3) \times U(1)_{\text{em}}, \quad (\text{D3})$$

or the corresponding colourless electroweak endpoints. The middle-dimensional data required at a DeVries point are

$$\mathfrak{I}_J(t_\star) = (\mathcal{B}(t_\star), \mathcal{H}_J(t_\star), K_J(t_\star, \lambda), \mathcal{R}_{\text{pole}}), \quad (\text{D4})$$

where $\mathcal{B}(t_\star)$ is the compact, boundary, or singular object, $\mathcal{H}_J(t_\star) = \text{span}\{h_J, a_J\}$ is the light order-parameter/current subspace, K_J is the reduced kernel, and $\mathcal{R}_{\text{pole}}$ is the matching rule to the transverse pole propagator. At the DeVries point the light two-channel reduction must take the form

$$\mathcal{O}_J(t_\star)|_{\mathcal{H}_J} \longrightarrow \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}. \quad (\text{D5})$$

The positive branch is then assigned to the pole-ratio target, and the negative branch is assigned to a scalar, boundary, brane, or compactification datum. The compatibility diagram

is

$$\begin{aligned}
(D, n_{\text{KK}}) &= (11, 7) \text{ or } (7, 3) \xrightarrow{t} (D, n_{\text{KK}}) = (9, 5) \text{ or } (5, 1), \\
(D, n_{\text{KK}}) &= (10, 6) \text{ or } (6, 2): \quad \mathcal{O}_J(t_\star)|_{\mathcal{H}_J} \rightarrow K_J(\lambda), \\
K_J(\lambda) &\longrightarrow (J_W, J_Z) = (J_H, J_{\text{adj}}), \\
(J_W, J_Z) &\longrightarrow \Delta_{V,T}^{-1}(s_{W,Z}).
\end{aligned} \tag{D6}$$

Thus the dimensional interpolation can contribute to the paper only through a kernel derivation, an ordered assignment theorem, and a pole-matching theorem.

4. Top-quark scale

The dimensional interpolation also creates a top-sector question. In the Standard Model, fermion masses are generated by Yukawa couplings to the Higgs vacuum expectation value. The Higgs coupling to a fermion is proportional to its mass, so the top quark has the largest direct fermionic coupling to the electroweak order parameter. This fact is visible in Higgs phenomenology and in the role of top loops in Higgs production, electroweak corrections, and vacuum stability [5, 8, 9].

The DeVries question is sharper: does the six-dimensional interior sector select the top quark as the fermion attached most directly to the order parameter? A positive answer would require a source-theory map

$$\mathcal{F}_{\text{top}} : \mathcal{H}_J \longrightarrow y_t, \quad m_t, \quad \text{or a top-sector boundary datum,} \tag{D7}$$

compatible with the same determinant that gives the W/Z pole-ratio clue. This is an open target. The present manuscript records the source-backed Standard Model fact and the required form of a DeVries explanation.

-
- [1] A. Rivero, “Mass terms to break susy-like degeneration,” arXiv:hep-ph/0606171 (2006).
 - [2] D. Tong, “String Theory,” University of Cambridge Part III lecture notes (2009).
 - [3] J. Polchinski, “What is String Theory?,” arXiv:hep-th/9411028 (1994).
 - [4] D. Tong, “The Standard Model,” University of Cambridge Part III lecture notes.
 - [5] L. Alvarez-Gaume and M. A. Vazquez-Mozo, “An Invitation to Quantum Field Theory and the Standard Model,” arXiv:2306.08097 [hep-th] (2023).

- [6] D. Tong, “Gauge Theory,” University of Cambridge lecture notes (2018).
- [7] C. Csaki, J. Hubisz, and P. Meade, “TASI Lectures on Electroweak Symmetry Breaking from Extra Dimensions,” arXiv:hep-ph/0510275 (2005).
- [8] S. Dawson, “TASI 2016 Lectures: Electroweak Symmetry Breaking and Effective Field Theory,” arXiv:1712.07232 [hep-ph] (2018).
- [9] H. E. Logan, “TASI 2013 lectures on Higgs physics within and beyond the Standard Model,” arXiv:1406.1786 [hep-ph] (2022).
- [10] S. P. Martin and D. G. Robertson, “Standard Model parameters in the tadpole-free pure MS scheme,” arXiv:1907.02500 [hep-ph] (2025).
- [11] T. Aaltonen *et al.* (CDF Collaboration), “High-precision measurement of the W boson mass with the CDF II detector,” Science **376**, 170–176 (2022).
- [12] E. Witten, “Search for a Realistic Kaluza-Klein Theory,” Nucl. Phys. B **186**, 412–428 (1981).
- [13] E. Witten, “Anomaly Cancellation On G2-Manifolds,” arXiv:hep-th/0108165 (2001).
- [14] B. Acharya and E. Witten, “Chiral Fermions from Manifolds Of G2 Holonomy,” arXiv:hep-th/0109152 (2001).
- [15] B. S. Acharya and S. Gukov, “M theory and Singularities of Exceptional Holonomy Manifolds,” arXiv:hep-th/0409191 (2004).
- [16] A. P. Braun, S. Cizel, M. Huebner, and S. Schafer-Nameki, “Higgs Bundles for M-theory on G2-Manifolds,” arXiv:1812.06072 [hep-th] (2019).
- [17] D. Tong, “Line Operators in the Standard Model,” arXiv:1705.01853 [hep-th] (2017).
- [18] P.-S. Hsin and J. Gomis, “Detecting Standard Model Gauge Group from Generalized Fractional Quantum Hall Effect,” arXiv:2411.18160 [hep-th] (2024).
- [19] J. Baez and J. Huerta, “The Algebra of Grand Unified Theories,” arXiv:0904.1556 [hep-th] (2010).
- [20] V. A. Britto, “A Mathematical Construction of an E6 Grand Unified Theory,” arXiv:2102.13465 [hep-th] (2021).
- [21] A. Strominger, “The Heterotic String is a Soliton,” arXiv:hep-th/9504047 (1995).
- [22] A. Sen, “String String Duality Conjecture in Six Dimensions and Charged Solitonic Strings,” arXiv:hep-th/9504027 (1995).
- [23] E. Witten, “Small Instantons in String Theory,” arXiv:hep-th/9511030 (1995).
- [24] V. Kumar and W. Taylor, “String Universality in Six Dimensions,” arXiv:0906.0987 [hep-th]

(2009).